Implicit Representations of 3D Geometry: A Comprehensive Survey, Benchmark, and Future Directions

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Implicit representations encode 3D geometry via field functions and associated geometric constraints that jointly characterize the target shape, naturally supporting gradient-based optimization and offering topological flexibility. Existing surveys adopt either a task-oriented or a network-oriented perspective, leaving the relationships among field design, surface extraction, and downstream deployment insufficiently systematized. This gap makes it difficult for researchers entering geometry modeling to form a structured understanding of implicit representations and their practical trade-offs. This survey provides a unified, representation-centric treatment of implicit 3D geometry, tracing the complete pipeline from field formulation to explicit surface extraction and downstream applications. We first formalize major implicit field families, including occupancy fields, signed distance fields, and unsigned distance fields, and their vector-, ray-, segment-, and medial-variant extensions, under a common abstraction defined by query domain, value domain, and geometry condition. This abstraction clarifies how different fields trade off expressiveness, regularity, and compatibility with downstream tasks. We then review iso-surface extraction algorithms that convert implicit fields to explicit meshes, followed by two representative applications: surface reconstruction and 3D generation. To support a fair and reproducible evaluation, we further introduce a unified benchmarking platform with a standardized interface covering 19 representative methods, enabling consistent evaluation of existing approaches and straightforward integration of future ones. The benchmark provides empirical findings on reconstruction accuracy, robustness, and efficiency. Finally, we discuss open challenges, future directions, and other applications of implicit 3D geometry.

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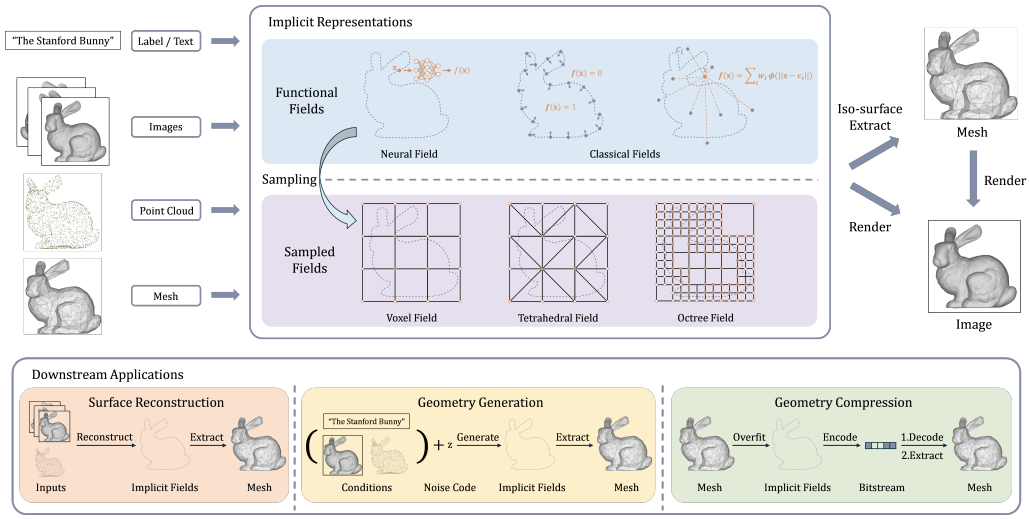


Fig. 1. Overview of the implicit representations.

1 INTRODUCTION

The rapid proliferation of digital twin technologies [61, 244], autonomous robotics [15, 106], immersive communication [188, 107], and spatial computing [169, 93] has propelled 3D geometry data to the forefront of modern computer science. As the bridge between physical reality and virtual simulation, 3D geometry data is indispensable to high-stakes applications ranging from medical imaging [170] and heritage preservation [110] to the generative pipelines of world models [74]. The choice of geometric representation governs the fidelity, efficiency, and flexibility of all downstream algorithms. Explicit representations specify the surface directly. The most common are polygon meshes, the standard output across geometry processing pipelines, which offer efficient rendering and manipulation. Other examples include point clouds, valued for flexible acquisition, and parametric surfaces such as NURBS [197], which provide mathematically precise, smooth definitions central to computer-aided design. However, the geometric detail of meshes and point clouds is inherently limited by the number of primitives, while parametric surfaces require tessellation for rendering and are difficult to fit to arbitrary topologies. Meshes in particular pose an additional challenge for modern learning-based pipelines: although their vertex positions can be optimized via gradient descent [180], their discrete combinatorial connectivity cannot be straightforwardly differentiated, preventing topology changes from emerging naturally during optimization.

Implicit representations take a fundamentally different approach. Rather than specifying the surface directly, they define a field over a spatial domain, within which the target geometry is characterized through conditions on that field, such as a level set. Different choices of the field function and the geometry condition yield representations with distinct trade-offs in expressiveness, regularity, and compatibility with downstream algorithms. Three principal scalar fields, the occupancy field (OF), the signed distance field (SDF), and the unsigned distance field (UDF), form the most widely adopted choices. Beyond them, varying the query domain, value domain, or geometry condition gives rise to vector, ray, segment, and medial representations.

Implicit fields share properties that make them naturally complementary to explicit meshes (Fig. 1). They admit gradient-based construction from diverse inputs while freely accommodating topological changes without remeshing. The resulting field can then be converted to explicit meshes

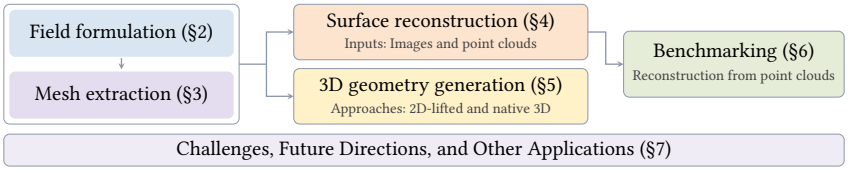


Fig. 2. Organization of this survey.

via iso-surface extraction (Sec. 3), bridging optimization-friendly implicit representations to the explicit meshes widely adopted in downstream geometry tasks.

Centering on this complementarity, this survey proceeds as outlined in Fig. 2. We first detail each field type under a unified framework (Sec. 2), then review iso-surface extraction algorithms (Sec. 3), followed by two major applications that have motivated a wide range of implicit methods: surface reconstruction from various kinds of inputs (Sec. 4) and 3D geometry generation (Sec. 5). We further benchmark point cloud surface reconstruction quantitatively, the longest-standing application of implicit fields (Sec. 6). Finally, we highlight open challenges and promising research directions that point beyond the current landscape of implicit representations and surface extraction. Note that two existing surveys are closely related to ours. Huang et al. [81] also review and benchmark point cloud surface reconstruction, but from a task-oriented perspective. Xie et al. [271] organize neural fields around network-level design choices such as conditioning, architecture, and forward mapping, adopting a network-oriented perspective. Ours takes a representation-centric view, extending beyond point cloud reconstruction to also cover image-based reconstruction and 3D geometry generation, and our benchmark additionally evaluates non-watertight geometry.

Throughout this survey, f denotes the field function in an implicit representation, including the occupancy function $f_{\mathrm{O}} : \mathbb{R}^3 \to \{0, 1\}$, the signed distance function $f_{\mathrm{S}} : \mathbb{R}^3 \to \mathbb{R}$, and the unsigned distance function $f_{\mathrm{U}} : \mathbb{R}^3 \to \mathbb{R}_{\geq 0}$; ∇f denotes its spatial gradient, while $\mathbf{S}$ and $\mathbf{M}$ denote the underlying surface and medial axis, respectively.

2 IMPLICIT REPRESENTATIONS

We formalize an implicit representation as a pair $\mathcal{F} = \big(f\colon \mathcal{Q} \to \mathcal{V},\ C\big)$, where f is a field function mapping a query domain $\mathcal{Q}$ to a value domain $\mathcal{V}$, and C defines how the target geometry $\mathcal{G} = C(f)$ is implicitly characterized by f, such as a level set of f. Different choices of f and C lead to representations with distinct trade-offs in expressiveness, regularity, and compatibility with downstream algorithms. Table 1 provides a systematic comparison, and Fig. 3 traces their chronological development. Following common usage, we use "implicit field" to refer to the representation $\mathcal{F}$ when no ambiguity arises; f denotes the underlying field function.

The field function f can be realized in two broad forms. *Functional forms* define f through the parameters of a chosen model; field values are obtained by evaluating the model at query points. Within this category, we further distinguish *classical* and *neural* models. Classical models define f through prescribed mathematical formulations, such as radial basis functions (RBFs) [82, 19, 245] and moving least squares (MLS) [98, 222, 2], whereas neural models parameterize f with learnable networks [238, 227, 56, 216, 141, 282]. Neural–discrete hybrids further couple discrete spatial structures storing local features with neural decoders for efficiency [237, 175, 241]. Functional forms offer flexibility through model selection and parameter fitting: field values and derivatives can be fitted at query points, and differential constraints such as the Eikonal condition can be incorporated into optimization. They can also be parameter-efficient and inherit regularity from the underlying model. *Sampled forms* instead store field values directly at finite spatial locations such as grid vertices [164, 41, 58]. Through cell lookup and local interpolation of the stored values, they

Table 1. Comparison of Implicit Geometry Representations.

	Occupancy Field (OF)	Signed Distance Field (SDF)	Unsigned Distance Field (UDF)	Vector Field (VF/NVF)	Ray Field (RF)	Segment Field (SF)	Medial Field (MF)
Definition	Binary inside/outside indicator	Signed Euclidean distance to surface	Unsigned Euclidean distance to surface	Vector toward nearest surface point	Distance along ray to first intersection	Segment–surface intersection indicator	Inscribed ball radius at nearest medial axis point
Field Formulation	$f_O : \mathbb{R}^3 \to \{0,1\}$	$f_S : \mathbb{R}^3 \to \mathbb{R}$	$f_U : \mathbb{R}^3 \to \mathbb{R}_{\geq 0}$	$f_V : \mathbb{R}^3 \to \mathbb{S}^2$ (VF) $f_N : \mathbb{R}^3 \to \mathbb{R}^3$ (NVF)	$f_R : \mathbb{R}^3 \times \mathcal{D} \to \mathbb{R}_{\geq 0}$	$f_{Seg} : \mathbb{R}^3 \times \mathbb{R}^3 \to \{0,1\}$ (+ $k_{Seg} : \mathbb{R}^3 \times \mathbb{R}^3 \to [0,1]$)	$f_M : \mathbb{R}^3 \setminus \mathbf{S} \to \mathbb{R}_{>0}$
Geometry Condition	$\partial\{f_O = 1\}$	$\{f_S = 0\}$	$\{f_U = 0\}$	$\{D_\Phi(f_V) = -1\}$ $\{f_N(\mathbf{x}) = 0\}$	$\{\mathbf{o} + f_R \cdot \mathbf{d} \mid f_R < +\infty\}$	$\{\mathbf{a} + k(\mathbf{b}-\mathbf{a}) \mid f_{Seg} = 1\}$	$\{f_M = \lvert f_S \rvert\}$ (gives **M**, not **S**)
Representation Capability	Closed, 2-manifolds only	Closed, 2-manifolds only	Arbitrary topology	Arbitrary topology	Arbitrary topology	Arbitrary topology	Closed, 2-manifolds only
Continuity	Discont. on **S** (relaxed: cont.)	Cont. everywhere	Cont. everywhere	f_V discont. on **S** & **M**; f_N discont. on **M**	Discont. where first-hit surface switches or vanishes	f_{Seg} discont. on $\partial\{f_{Seg} = 1\}$; (k cont. where $f_{Seg} = 1$)	Undef. on **S**; discont. across **S**
Differentiability	Non-diff. on **S**	Non-diff. on Γ & **M**	Non-diff. on **S** & **M**	Non-diff. exactly where discont.	Non-diff. exactly where discont.	Non-diff. exactly where discont.	Undef. on **S**; non-diff. on **M**
Field Properties	No metric information	∇f_S = outward $\mathbf{n}$; $\lVert \nabla f_S \rVert = 1$ (off Γ, **M**); 1-Lipschitz	$\nabla f_U \parallel \mathbf{n}$; $\lVert \nabla f_U \rVert = 1$ (off **S**, **M**); 1-Lipschitz	$f_V \parallel \mathbf{n}$; $f_V = -\nabla f_U$; $f_N \parallel \mathbf{n}$; $\lVert f_N \rVert = f_U$	$f_U = \min_{\mathbf{d}} f_R$; $\nabla_{\mathbf{o}} f_R \cdot \mathbf{d} = -1$	$\nabla_{\mathbf{a}} k \parallel \mathbf{n}$; $\nabla_{\mathbf{b}} k \parallel \mathbf{n}$	$f_M \geq \lvert f_S \rvert$, $=$ on **M**; $\nabla f_M \perp \nabla f_S$
Geometric Queries	Inside/outside	Signed distance; sphere tracing; surf. proj. with normal	Unsigned distance; surf. proj. with normal	VF: normal direction; NVF: O(1) surf. proj.	First-hit depth and point	Edge intersection test & point	Medial sphere tracing; O(1) medial axis proj.; local thickness
Edge–Surface Test	$(f_O - \tau)$ sign change	f_S sign change	∇f_U reversal	f_V / f_N direction reversal	Ray hit within edge	Direct query f_{Seg}	$\nabla(f_M - \lvert f_S \rvert)$ reversal

also support efficient evaluation at arbitrary query points. Adaptive structures such as octrees [58] and sparse grids [247] alleviate memory cost by storing samples selectively rather than on a dense uniform grid. However, even with exact stored samples, the accuracy of the interpolated field and its gradients remains limited by sampling resolution and the interpolation rule. Coarse sampling can introduce substantial errors, particularly in gradients, making differential properties such as the Eikonal condition difficult to preserve. The two forms are complementary: a functional form can always be sampled onto a discrete grid to yield its sampled counterpart, which provides a standard input representation for isosurface-extraction algorithms such as Marching Cubes.

Regardless of its form, f can be obtained from existing geometry or inferred from raw observations. For existing geometry, meshes support direct field evaluation, while skeleton-based methods derive fields from skeletal primitives via distance surfaces [208] or convolution surfaces [12]. Inference from observations relies on complementary mechanisms for constraining f. *Field formulations* prescribe f as the solution of a mathematical or physics-inspired model that builds structural properties into the field definition itself, spanning interpolation and approximation energies [19, 98], numerical Poisson solvers [90, 91, 195, 72], potential-theoretic formulations including charge-distribution potentials [193] and double-layer potentials [85, 7, 126, 276], and phase-field models [127]. *Data fidelity* anchors f to measurements such as point samples, oriented normals, or images via differentiable rendering. *Field regularization* imposes differential constraints on f such as the Eikonal condition, divergence penalties, and higher-order structure [62, 9, 280]. *Shape priors* inject geometric knowledge beyond the current input, from hand-designed geometric initialization [9] to data-driven priors learned from training data [191, 167].

The remainder of this section examines specific field types. Secs. 2.1, 2.2, and 2.3 cover the three principal scalar fields, namely the occupancy field, the signed distance field, and the unsigned distance field, respectively, and Sec. 2.4 broadens the scope to vector, ray, segment, and medial variants.

2.1 Occupancy Fields

The occupancy field (OF) defines a binary indicator function over $\mathbb{R}^3$, assigning 1 to points inside a shape and 0 otherwise; the surface is given by the boundary of the occupied region:

$$f_O : \mathbb{R}^3 \to \{0,1\}, \quad C : \mathbf{S} = \partial\{\mathbf{x} \in \mathbb{R}^3 \mid f_O(\mathbf{x}) = 1\}.$$

In practice, this discontinuous indicator is often approximated using neural or classical models, with the surface extracted by thresholding the resulting field. Neural models parameterize a continuous occupancy probability $p_\theta(\mathbf{x}) \in [0,1]$ [167, 30]. Among classical models, Poisson formulations

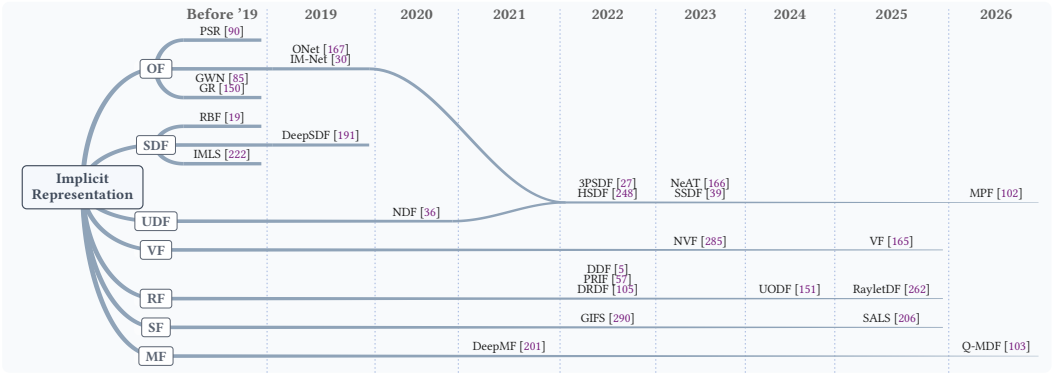


Fig. 3. Evolution of implicit representations.

define an indicator approximation through a least-squares fit of its gradient to a vector field derived from oriented samples [90]. Generalized winding number (GWN) and related Gauss formulations express indicator-like fields as sums of contributions from oriented triangles or points [85, 150, 7]. The regularized GWN (Reg-GWN) smooths the singular kernel of the point-cloud winding-number formulation, yielding a differentiable field [24].

Notably, OF inherently assumes *closed, 2-manifold* surfaces, as the inside–outside labeling is ill-defined for open boundaries or non-manifold edges (e.g. three or more faces sharing an edge). Moreover, it lacks metric distance information: its values saturate away from the surface, leading to vanishing gradients and limiting geometry-aware operations such as sphere tracing and nearest-point queries.

2.2 Signed Distance Fields

The signed distance field (SDF) is a scalar field over $\mathbb{R}^3$ whose value at each point equals the signed Euclidean distance to the nearest surface point, with negative values inside and positive values outside the shape:

$$f_S : \mathbb{R}^3 \to \mathbb{R}, \quad C : \ S = \{ \mathbf{x} \in \mathbb{R}^3 \mid f_S(\mathbf{x}) = 0 \}.$$

Similar to OF, SDF requires a *closed, 2-manifold* surface to define the sign, but further encodes metric distance information, giving rise to useful field properties and geometric queries.

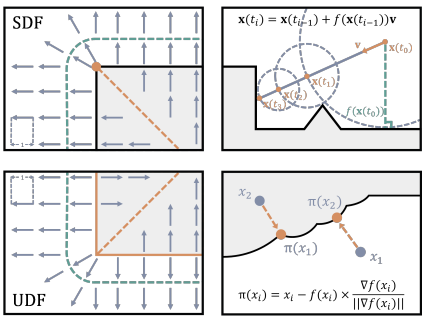


Fig. 4. Properties and applications of distance fields. Left: field properties of SDFs and UDFs: solid curves denote surface boundaries, blue arrows indicate unit gradients, green dashed curves denote offset surfaces, and orange marks non-differentiable loci, including sharp features, medial axes shown as dashed curves, and UDF surfaces shown as solid curves. Right: geometric queries, including sphere tracing and closest-point projection.

Field properties. Wherever differentiable, f_S satisfies the Eikonal equation $\|\nabla f_S\| = 1$, with the gradient equal to the outward unit normal $\mathbf{n}$ at the closest surface point. As illustrated in Fig. 4 (top left), this yields a gradient field (blue) that is continuous across the surface, with level sets (green) forming uniformly spaced offset surfaces. The field is non-differentiable at two types of loci: the medial axis $\mathbf{M}$ (orange dashed), where the closest surface point is non-unique and hence the gradient is undefined; and the sharp feature set $\Gamma \subset \mathbf{S}$ (orange dot), such as edges and corners, where the surface normal is undefined. The Eikonal property implies that f_S is globally 1-Lipschitz, i.e., $|f_S(\mathbf{x}) - f_S(\mathbf{y})| \leq \|\mathbf{x} - \mathbf{y}\|$. Enforcing $\|\nabla f_S\| \approx 1$ via Eikonal regularization has become standard practice in neural SDF optimization [62].

Geometric queries. The 1-Lipschitz property guarantees that $|f_S(\mathbf{x})|$ is a conservative lower bound on the distance to S, enabling sphere tracing [67]: as shown in Fig. 4 (top right), a ray starting from $\mathbf{x}(t_0)$ along direction $\mathbf{v}$ is safely advanced by $\mathbf{x}(t_i) = \mathbf{x}(t_{i-1}) + f(\mathbf{x}(t_{i-1}))\,\mathbf{v}$ without overshooting. Sphere tracing terminates when $|f_S| < \tau$ for a small threshold τ. When neural approximation errors cause overshooting, a sign change between consecutive steps can bracket the crossing for refinement. This technique is widely adopted in differentiable rendering pipelines [87, 134]; architectures that enforce Lipschitz continuity by design, such as 1-LNDF [40], further guarantee robustness. Additionally, the closest surface point can be recovered via projection $\Pi(\mathbf{x}) = \mathbf{x} - f_S(\mathbf{x})\,\nabla f_S(\mathbf{x})/\|\nabla f_S(\mathbf{x})\|$(Fig. 4, bottom right), which is widely used to formulate geometric losses for fitting SDFs to raw point clouds [155].

A practical variant, the truncated SDF (TSDF), clamps values to a narrow band $f_T(\mathbf{x}) = \mathrm{clip}(f_S(\mathbf{x}), -\delta, \delta)$, sacrificing global distance outside the truncation region but concentrating representation capacity near the surface and improving robustness to noise. It is widely adopted in real-time volumetric reconstruction and SLAM [41, 178] as well as in 3D geometry compression [240, 239, 205].

2.3 Unsigned Distance Fields

The unsigned distance field (UDF) is a scalar field over $\mathbb{R}^3$ whose value at each point equals the unsigned Euclidean distance to the nearest surface point:

$$f_U : \mathbb{R}^3 \to \mathbb{R}_{\geq 0}, \quad C : \mathrm{S} = \{\, \mathbf{x} \in \mathbb{R}^3 \;\big|\; f_U(\mathbf{x}) = 0 \,\}.$$

Unlike OF and SDF, UDF requires no global inside/outside classification, enabling it to represent surfaces of arbitrary topology, including open and non-manifold geometry.

Field properties. UDF shares several properties with SDF: it satisfies the Eikonal condition $\|\nabla f_U\| = 1$ wherever differentiable and is globally 1-Lipschitz, with the gradient parallel to the surface normal at the nearest point. Fig. 4 reveals the key structural difference between UDF and SDF: the UDF gradient (blue) points away from S on both sides and reverses discontinuously across the surface. Thus, UDF is non-differentiable on the surface, in addition to the medial axis (orange).

Geometric queries. The closest surface point can also be recovered via projection $\Pi(\mathbf{x}) = \mathbf{x} - f_U(\mathbf{x})\,\nabla f_U(\mathbf{x})/\|\nabla f_U(\mathbf{x})\|$ (Fig. 4, bottom right). Sphere tracing also applies to UDFs, with termination when $f_U < \tau$. However, neural approximation errors may cause overshooting without a sign change to bracket the crossing, making intersection detection less reliable.

Challenges. The non-differentiability at the zero-level set poses difficulties for both learning and surface extraction. On the learning side, neural approximations inevitably smooth the sharp minimum at S, corrupting distance values and gradients near the surface [296, 54] and producing fragmented reconstructions. On the extraction side, standard iso-surface algorithms such as Marching Cubes [149] rely on sign changes between adjacent grid vertices; since UDF is strictly non-negative, such detection fails, necessitating specialized algorithms that identify the zero-level set without sign flips.

2.4 Other Implicit Representations

For a watertight and manifold shape, the three scalar fields above are closely related (Fig. 5): OF can be obtained by thresholding SDF, $f_O(\mathbf{x}) = \mathbf{1}[f_S(\mathbf{x}) < 0]$, discarding all metric distance information; UDF is the absolute value of SDF, $f_U(\mathbf{x}) = |f_S(\mathbf{x})|$, removing the sign at the cost of non-differentiability at the zero-level set. Beyond these scalar fields, representations have

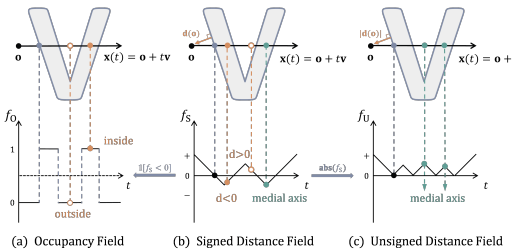


Fig. 5. Visual illustrations of OF, SDF, and UDF in 2D.

been proposed that enrich the value domain to vectors, change the query domain from points to rays or segments, or target skeletal geometry rather than the surface. We survey each family in turn below.

2.4.1 Vector fields. Vector fields (Fig. 6(a)) replace scalar outputs with 3D vectors pointing toward the nearest surface point, inheriting UDF's ability to represent arbitrary topology while additionally offering direct normal and projection queries.

VF [165] predicts only the direction toward the closest surface point:

$$f_{\mathrm{V}}\colon \mathbb{R}^3 \to \mathbb{S}^2, \quad C:\ \mathbf{S} = \{\, \mathbf{x} \in \mathbb{R}^3 \;\big|\; D_\Phi(f_{\mathrm{V}})(\mathbf{x}) = -1 \,\},$$

where D_Φ denotes the flux-density operator. Since $f_{\mathrm{V}} = -\nabla f_{\mathrm{U}}$ away from $\mathbf{S}$ and $\mathbf{M}$, it directly encodes the surface normal. For meshing, voxels whose discrete flux density falls below a threshold are flagged as containing $\mathbf{S}$; the eight predicted directions at each flagged voxel's vertices are then split into two clusters by cosine similarity to assign inside–outside labels for standard iso-surface extraction.

NVF [285] predicts the full displacement toward the closest surface point:

$$f_{\mathrm{N}}\colon \mathbb{R}^3 \to \mathbb{R}^3, \quad C:\ \mathbf{S} = \{\, \mathbf{x} \in \mathbb{R}^3 \;\big|\; f_{\mathrm{N}}(\mathbf{x}) = \mathbf{0} \,\},$$

whose norm equals f_{U} away from $\mathbf{M}$ and whose output yields $\mathrm{O}(1)$ surface projection via $\mathbf{x} + f_{\mathrm{N}}(\mathbf{x})$ in a single, differentiation-free forward pass.

2.4.2 Ray fields. Ray fields (Fig. 6(b)) lift the input from a single point to an oriented ray $(\mathbf{o}, \mathbf{d})$. In its canonical form, a ray field encodes the distance along the ray to its first surface intersection:

$$f_{\mathrm{R}}\colon \mathbb{R}^3 \times \mathcal{D} \to \mathbb{R}_{\geq 0}, \quad C:\ \mathcal{S} = \left\{\, \mathbf{o} + f_{\mathrm{R}}(\mathbf{o}, \mathbf{d})\, \mathbf{d} \;\middle|\; f_{\mathrm{R}}(\mathbf{o}, \mathbf{d}) < +\infty \right\}.$$

This recovers a surface point with a single query per ray. Since $f_{\mathrm{U}}(\mathbf{o}) = \min_{\mathbf{d}} f_{\mathrm{R}}(\mathbf{o}, \mathbf{d})$, ray fields generalize UDF while providing richer directional information; the spatial gradient further satisfies a directed eikonal property $\nabla_{\mathbf{o}} f_{\mathrm{R}} \cdot \mathbf{d} = -1$ [5]. However, f_{R} is inherently discontinuous: small perturbations in position or direction can switch the first-hit surface, and its directional dependence precludes standard iso-surface extraction.

DDF [5] and RayDF [145] retain the full $\mathcal{D} = \mathbb{S}^2$ domain: the former predicts multiple candidate depths per ray with learned blending weights to smooth over occlusion boundaries, while the latter constrains different rays hitting the same surface point to agree in 3D. Operating on camera rays, DRDF [105] redefines the output as signed along the ray so that the field transitions continuously through zero at the surface, enabling robust extraction under prediction uncertainty, while PRIF [57] reparameterizes each ray by its perpendicular foot to the origin and directly outputs the 3D hit point, ensuring a unique input for each geometric ray. UODF [151] restricts distance queries to three orthogonal directions, enabling direct surface-point estimation along each axis and reducing interpolation errors from scalar distance values. RayletDF [262] further localizes the representation by replacing global rays with short segments originating near the surface and blending predictions via learned confidence for cross-scene generalization.

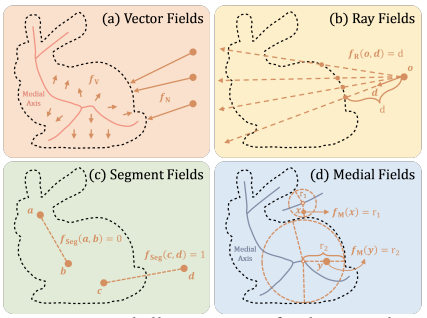


Fig. 6. Visual illustration of other implicit fields in 2D.

2.4.3 Segment fields. Segment fields (Fig. 6(c)) take a line segment rather than a single point or ray as input. They are characterized by an intersection indicator and an interpolation parameter:

$$f_{\mathrm{Seg}}\colon \mathbb{R}^3 \times \mathbb{R}^3 \to \{0, 1\}, \qquad k\colon \mathbb{R}^3 \times \mathbb{R}^3 \to [0, 1],$$

where $f_{\text{Seg}}(\mathbf{a}, \mathbf{b}) = 1$ indicates that the segment from $\mathbf{a}$ to $\mathbf{b}$ intersects the surface, and $k(\mathbf{a}, \mathbf{b})$ localizes the intersection along the segment. The surface is then

$$C\colon \mathbf{S} = \left\{ \mathbf{a} + k(\mathbf{a}, \mathbf{b})\,(\mathbf{b} - \mathbf{a}) \;\middle|\; f_{\text{Seg}}(\mathbf{a}, \mathbf{b}) = 1 \right\}.$$

They handle arbitrary topology and detect edge–surface intersections by a direct query of f_{Seg}, requiring no sign change or gradient reversal detection.

GIFS [290] infers the intersection parameter k from a jointly learned UDF, whereas SALS [206] directly predicts both the indicator and k in a single forward pass, further showing that $\nabla_{\mathbf{a}} k$ and $\nabla_{\mathbf{b}} k$ are parallel to the surface normal, enabling normal recovery without additional queries.

2.4.4 Medial fields. The medial field (Fig. 6(d)) targets skeletal geometry rather than the surface itself, building on the medial axis transform [13, 236]:

$$f_{\text{M}}\colon \mathbb{R}^3 \setminus \mathbf{S} \to \mathbb{R}_{>0},\; C:\; \mathbf{M} = \{\, \mathbf{x} \in \mathbb{R}^3 \setminus \mathbf{S} \;|\; f_{\text{M}}(\mathbf{x}) = |f_{\text{S}}(\mathbf{x})| \,\},$$

where $f_{\text{M}}(\mathbf{x})$ equals the length of the medial spoke through $\mathbf{x}$, connecting its nearest surface point to the medial axis. It is undefined on $\mathbf{S}$, equals $|f_{\text{S}}|$ exactly on $\mathbf{M}$, and is larger elsewhere.

Learned jointly with SDF [201], the medial field encodes local thickness and enables $O(1)$ projection onto $\mathbf{M}$ as well as medial sphere tracing, which improves convergence over standard sphere tracing. Since f_{M} is derived from SDF, it is also restricted to *closed, 2-manifold* shapes.

3 IMPLICIT FIELDS TO EXPLICIT MESHES

Given an implicit field, surface extraction produces an explicit mesh approximating the underlying geometry, bridging implicit fields to deployable meshes for rendering, simulation, and geometry processing. We focus on OF, SDF, and UDF, which have motivated the richest body of extraction algorithms; other field types either reuse similar machinery or recover surface points directly from their field queries before applying downstream meshing (Sec. 2.4).

We organize methods by field type: OF and SDF admit classical isosurfacing, while UDF requires specialized algorithms due to the absence of sign information. For each, we first describe general extraction algorithms, then their differentiable and learning-based variants.

3.1 Extraction from OF and SDF

OF and SDF both induce an inside/outside partition of space, and the surface is defined as the boundary between the two regions: the zero level set for SDF and an occupancy iso-surface for OF. SDF additionally encodes metric distance, enabling methods that exploit this richer information.

3.1.1 General Extraction Algorithms. We group methods into sign-based and metric-based, both operating on a sampled grid, and efficient strategies tailored specifically to neural fields.

Sign-based methods apply to any field that provides an inside/outside classification, including both OF and SDF. They sample the field on a grid and identify cells whose vertices exhibit mixed signs or occupancy values, indicating that the surface passes through them. They differ in how mesh vertices are placed and connected within or around these cells.

Marching Cubes (MC) [149] is the most widely used iso-surfacing algorithm. It classifies each vertex of a regular grid as inside or outside, and a precomputed lookup table maps the 2^8 possible sign configurations per cell to triangle patches, with vertex positions linearly interpolated from the field values along sign-changing edges.

MC is efficient and embarrassingly parallel, but (i) vertices are confined to grid edges, limiting fidelity for sharp features (Fig. 7, MC); and (ii) topological ambiguities arise when diagonally opposite face vertices share the same sign (Fig. 7, MC (Ambiguous)), with analogous ambiguities inside the cell. The Asymptotic Decider [182] resolves face ambiguities via the bilinear interpolant at

the face saddle point; MC33 [34] extends this to interior ambiguities; and subsequent variants [148, 111] address numerical degeneracies. Marching Tetrahedra (MT) [47] sidesteps ambiguities by subdividing each cell into tetrahedra, for which linear interpolation yields a unique intersection pattern, at the cost of more triangles and directional bias from the chosen subdivision.

Dual Contouring (DC) [88] places one vertex per sign-changing cell by minimizing a Quadric Error Function (QEF) over Hermite data (edge–surface intersections with normals). For each sign-changing edge, the four adjacent dual vertices are connected into a non-planar quadrilateral, which can be split diagonally into triangles. Because the vertex is not constrained to grid edges, DC preserves sharp features better than MC, but the unconstrained minimizer can drift outside the cell and cause self-intersections, and the one-vertex-per-cell assumption produces non-manifold output when multiple sheets traverse a single cell (Fig. 7, DC).

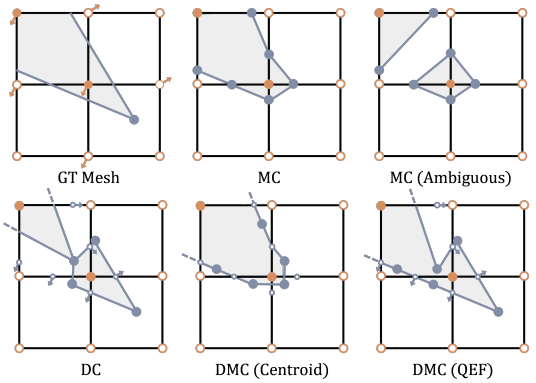


Fig. 7. Comparison of surface extraction methods in 2D. At grid vertices, filled and hollow orange dots indicate inside and outside samples, respectively, with arrows showing ground-truth normals. Hollow blue dots denote predicted edge–surface intersections; filled blue dots are the final mesh vertices. Gray shading depicts the reconstructed surface.

Dual Marching Cubes (DMC) [181, 217] mitigates both MC's edge-constrained vertex placement and DC's one-vertex-per-cell limitation by constructing a disambiguated MC patch graph that separates distinct sheets within each cell, placing one dual vertex per patch and stitching patches into a guaranteed manifold mesh. [181] positions each vertex at the patch centroid for robustness (Fig. 7, DMC (Centroid)), while [217] minimizes a QEF per patch to better preserve sharp features (Fig. 7, DMC (QEF)).

Metric-based methods exploit the distance information encoded by SDFs, beyond the inside/outside signs used by the methods above. For an exact SDF, each sample $\mathbf{x}$ defines an empty tangent ball centered at $\mathbf{x}$ with radius $|f_S(\mathbf{x})|$: its interior contains no surface, while its boundary touches $\mathbf{S}$ at a nearest point. This property is mainly exploited in two ways. First, the tangency condition of these spheres is used to recover surface contact points or formulate reconstruction objectives. Second, these radii are squared and used as weights to construct a power diagram that provides an SDF-aware spatial structure for contouring, filtering, and refinement. The empty-ball and tangency properties hold for exact SDF samples; inaccurate distances may violate these properties and compromise reconstruction.

Along the tangency-based route, RFTS [220] optimizes an initial mesh toward tangency with the distance spheres, while preserving its initial topology. RFTA [219] instead optimizes tangent points along the feasible arcs of the spheres, collects them into an oriented point cloud, and reconstructs the surface via Poisson Surface Reconstruction (PSR) [90]. MES [96] computes such contact points deterministically using Lie sphere geometry, avoiding the iterative optimization required by RFTA. DCSDD [20] incorporates the same tangency constraints into Dual Contouring, iteratively refining dual vertices and Hermite data from sampled SDF values to improve sharp-feature reconstruction.

Along the power-diagram route, Kohlbrenner et al. [97] organize the weighted SDF samples into a regular triangulation and its dual power diagram. The regular triangulation supports tetrahedral contouring, while interfaces between opposite-sign power cells define a Power Contour; the same structure also guides adaptive insertion of new SDF samples in insufficiently constrained regions. GCP [94] reuses the regular triangulation to approximate SDF gradients and generate surface contact points, then filters invalid projections using neighboring distance spheres before PSR. Wang

et al. [255] further augment the weighted sites with surface projections and adaptively insert new samples in regions where the current mesh deviates most from the underlying SDF, updating the power diagram incrementally.

Efficient extraction for neural fields. Evaluating a neural field on a dense grid incurs many redundant queries far from the surface. MISE [167] hierarchically refines only cells with sign changes via an octree, and McGrids [203] constructs an adaptive tetrahedral grid via Monte Carlo sampling near the surface, both substantially reducing network evaluations. For ReLU MLPs, Analytic Marching [108] and Marching Neurons [231] avoid gridding by exploiting piecewise linearity: the zero set is planar within each linear region. The former marches among neighboring regions from a seed point but may miss disconnected components; the latter traverses the network layer by layer, splitting cells along neuron hyperplanes, naturally handling arbitrary topology.

3.1.2 Differentiable and Learning-Based Variants. Two strategies integrate mesh extraction into gradient-based pipelines: making an existing algorithm differentiable, or training a network to learn the extraction directly.

Differentiable extraction layers. MeshSDF [202] keeps standard MC in the forward pass and derives analytic gradients for extracted surface samples by modeling how SDF perturbations locally move the zero crossing. Recent differentiable variants operate on sampled fields and make mesh extraction optimization-friendly by parameterizing vertex placement within local cells. DMTet [223] applies this idea to MT on deformable tetrahedral grids, TetWeave [11] to Delaunay tetrahedralizations with directional signed distances, and FlexiCubes [224] to DMC by augmenting SDF values with auxiliary extraction parameters.

Learned extraction modules. Deep Marching Cubes [120] replaces discrete sign classification with continuous occupancy probabilities, making topology selection differentiable. Neural Marching Cubes [32] redesigns tessellation templates with additional intra-cell vertices and trains a 3D CNN to predict per-cell topology and vertex positions from local context, recovering sharp edges beyond classical MC's reach. Neural Dual Contouring [31] and Self-Supervised DC [235] replace DC's QEF solve with learned vertex prediction, the latter removing the need for ground-truth mesh supervision. A separate line of work replaces the fixed grid with Voronoi/Delaunay structures: VoroMesh [163] directly uses Voronoi bisector faces between oppositely labeled points as surface elements, while PoNQ [162] computes a quadric-optimal vertex per cell and connects them via Delaunay tetrahedralization with graph-cut labeling, achieving better sharp-feature preservation with fewer faces. Both are trained end-to-end on input SDF grids.

3.2 Extraction from UDF

UDF naturally supports open and non-manifold geometry that OF and SDF cannot represent. However, extraction is harder: the field is everywhere non-negative, lacking sign changes, and learned UDFs often fail to attain exact zero at the surface since smooth networks round off the non-differentiable minimum. Most methods therefore rely on UDF gradients as an additional cue, though recent power-diagram-based approaches localize the surface from the distance values alone.

3.2.1 General Extraction Algorithms. We group methods into four families: sign-inference methods, double-covering methods, dual-contouring-style methods, and power-diagram-based methods.

Sign-inference methods assign pseudo-signs to grid vertices using UDF gradients, then apply MC or its multi-label variant for extraction. MeshUDF [65] and CAP-UDF [303] select an anchor vertex per cell and compute gradient inner products with the remaining vertices to assign binary pseudo-signs; MeshUDF additionally enforces global consistency via breadth-first propagation. GeoUDF [204] instead evaluates gradient constraints for every vertex pair within each cell to jointly determine the sign configuration, selecting the best-matching MC template via a cost function.

MIND [29] handles non-manifold surfaces by first constructing local two-signed fields around manifold patches, then extending them into a global multi-labeled field for multi-labeled MC extraction. All four methods rely on UDF gradients, making sign assignment unreliable near **M** where neighboring vertices may have closest points on different surface sheets.

Double-covering methods extract a double-layered shell at a small positive offset via standard MC, project it onto the true surface via a covering map, and apply min-cut to remove one layer. The min-cut assumes an orientable manifold surface; non-manifold structures or non-orientable manifolds, such as a Möbius strip, cannot be faithfully recovered. DCUDF [73] introduces this pipeline; DCUDF2 [28] adds an accuracy-aware loss and topology editing to reduce offset sensitivity.

Dual-contouring-style methods partition space into local regions, optimize one representative surface vertex per region from local tangent constraints, and recover connectivity from region dual adjacency. DualMesh-UDF [296] instantiates this on an adaptive octree, solving a QEF in each non-empty leaf from tangent planes estimated from filtered off-surface samples. VoroUDF [251] replaces the octree with an optimized Voronoi partition, updates seed positions using local tangent constraints within each Voronoi cell and derives connectivity from approximate geodesic Voronoi adjacency, thereby better preserving open boundaries, sharp features, and non-manifold geometry.

Power-diagram-based methods extend the metric-based construction above to exact UDFs, where surface crossings must be identified without sign changes. IRCUdf [95] recursively tests whether edges of a regular triangulation are covered by the distance spheres; edges that may contain the zero set are then used for Dual Contouring. SpUDD [250] instead constructs a *superpower contour* from power faces outside all distance spheres, uses it to identify active grid edges, and applies an unsigned adaptation of DCSDD [20] to recover the final mesh.

3.2.2 Learning-Based Variants. NSD [229] trains a lightweight MLP to predict per-cell sign configurations from UDF values and gradients, converting the UDF locally into a pseudo-SDF meshable by standard MC; its extension [228] adds iterative passes propagating neighborhood context to correct ambiguous cells. UNDC [31] adapts its learned DC framework to UDF, predicting dual vertices and edge occupancy from UDF values alone.

4 SURFACE RECONSTRUCTION

Recovering continuous surfaces from sensor observations such as images or point clouds is essential for downstream rendering, simulation, and editing. As discussed in earlier sections, implicit fields serve as a natural bridge in this pipeline: they can be reconstructed from diverse observation types and then converted to explicit meshes via iso-surface extraction.

We focus on deterministic methods that seek the single field best explaining the input; generative approaches that model a distribution over plausible shapes are covered in Sec. 5. The rest of this section is organized by input modality: **2D images** and **3D point clouds**.

4.1 2D Images as Inputs

Reconstructing a surface from 2D images is attractive because images are easy to capture, yet the geometric constraints they provide are indirect, coupling geometry with illumination, texture, and specular effects. Implicit fields are well suited to this setting because they can be differentiably rendered, enabling direct supervision from image observations.

We discuss two task settings: *single-view reconstruction*, where a single photograph leaves the geometry severely under-constrained, and *multi-view reconstruction*, where cross-view consistency substantially constrains it. Fig. 8 summarizes the evolution of these methods.

4.1.1 Single-view Reconstruction. The methods reviewed below learn an encoder–decoder mapping from a single image to a neural implicit field: an encoder extracts image features that condition a

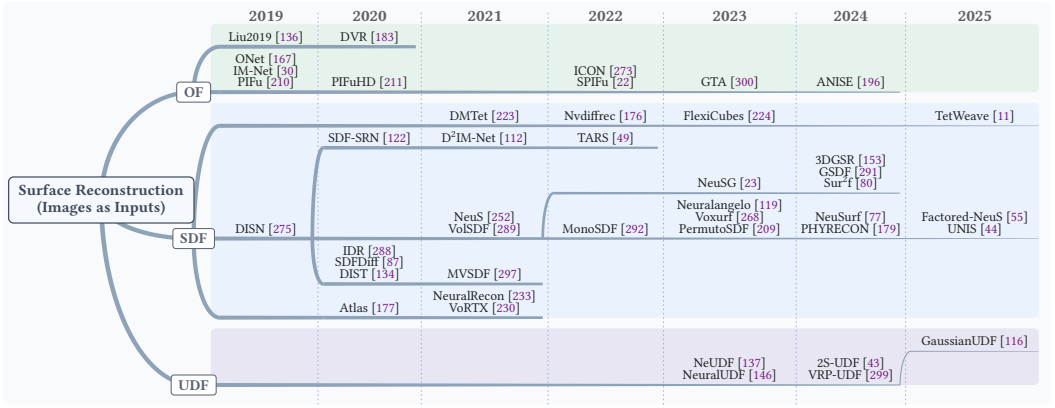


Fig. 8. Evolution of implicit surface reconstructions (2D images as inputs).

coordinate-based decoder defining the field. They differ primarily in supervision: some train with paired *3D ground truth*, while others rely solely on *2D observations*. In both cases, the learned prior enables plausible completion of occluded regions for categories seen during training.

Learning with 3D supervision. When paired 3D ground truth is available, the network is trained by directly minimizing the discrepancy between predicted and reference field values. The central design choice is then what image features the encoder extracts to condition the field. The earliest methods, such as ONet [167] and IM-Net [30], encode the image into a single global latent code, but this tends to oversmooth local detail. To recover finer geometry, PIFu [210] introduces pixel-aligned conditioning for clothed human reconstruction, projecting each 3D query into the image and sampling the corresponding local feature, with subsequent work enriching conditioning or injecting domain-specific priors [211, 300, 273, 22]. For general objects, DISN [275] combines global and local features and learns the 2D projection of each 3D query, removing the need for known camera parameters. Other strategies include coarse-to-fine refinement [112] and part-level assembly [196].

Learning with 2D supervision. When paired 3D supervision is unavailable, the network is instead trained through differentiable image formation: points sampled along camera rays are queried and the resulting predictions are compared against observed silhouettes or colors. Liu et al. [136] instantiate this idea for OFs under multi-view silhouette supervision by aggregating occupancy along each ray into a foreground probability. DVR [183] finds the first ray–surface intersection of the OF and evaluates a texture field there, deriving an analytic depth gradient via implicit differentiation so that both mask and RGB observations supervise geometry. SDF-SRN [122] switches to an SDF with SRN-style ray marching and replaces binary masks with silhouette distance transforms, making every pixel contribute geometric signal. TARS [49] extends this SDF-based rendering with a topology-aware deformation into a canonical space, enabling category-level correspondence learning under 2D supervision alone.

4.1.2 Multi-view Reconstruction. With multiple views available, the observed geometry is often well determined, and the focus shifts to how best to recover it. Most methods assume known camera poses, although some jointly refine or even estimate them during reconstruction. We group multi-view approaches into three families: *feed-forward prediction* from the input views, *per-instance optimization* that fits the field directly to the observed images, and *hybrid representations* that couple an implicit field with an explicit geometric proxy.

Feed-forward prediction. These methods fuse image features into a voxel grid to predict a TSDF in a single forward pass [177, 233, 230], differing mainly in the fusion strategy. They are fast and generalizable, but their fixed voxel resolution limits geometric fidelity.

Per-instance optimization. In contrast, per-instance methods fit an implicit field to each test instance so that its differentiable renderings explain the input views. Distance fields are usually preferred over OFs for their richer geometric signal. This per-instance fitting is slower than feed-forward prediction, but usually yields higher fidelity. The main difference then lies in how the implicit field is differentiably rendered into images.

Surface rendering supervises an SDF at the first ray–surface intersection, typically located by sphere tracing; methods differ mainly in how this intersection is differentiated. IDR [288] and MVSDF [297] backpropagate through the zero-level-set hit point via implicit differentiation. SDFDiff [87] restricts the gradient to the local grid neighborhood around the hit, while DIST [134] introduces a practical differentiable sphere-tracing layer for neural SDFs with efficient forward and backward approximations.

Volume rendering was popularized in neural rendering by NeRF [169], which represents a scene as a continuous radiance field that predicts view-independent density and view-dependent radiance and integrates their contributions along camera rays. However, unlike the implicit representations included in our taxonomy, NeRF's learned density does not explicitly define a surface: no particular density level set is prescribed as the geometry, and the field is constrained primarily through image reconstruction. Consequently, image-consistent density fields need not correspond to accurate geometry, and their extracted isosurfaces can be noisy and sensitive to the chosen threshold. SDF-based volume rendering addresses this ambiguity at the representation level by parameterizing geometry with an SDF, whose zero level set explicitly defines the surface. The SDF is then mapped to density or opacity weights for volume rendering, retaining NeRF-style differentiable image supervision while making the reconstructed surface explicit. Two concurrent works established this paradigm: VolSDF [289] maps the SDF to density with a Laplace CDF and derives an opacity approximation error bound to guide sampling, whereas NeuS [252] uses a logistic-based mapping and derives first-order unbiased rendering weights that peak exactly at the zero level set. UNIS [44] unifies these mappings in a broader family and characterizes the conditions for first-order unbiasedness. Follow-up work extends this paradigm, e.g., with explicit multi-view geometric constraints [59], stronger geometric priors for under-constrained settings [292, 77], more efficient spatial parameterizations [119, 209, 268], and richer objectives such as physical plausibility or inverse rendering [179, 55]. To handle non-watertight surfaces, UDF-based volume rendering extends this idea to unsigned distances, with the central challenge being their conversion into reliable ray weights. NeuralUDF [146] and NeUDF [137] initiated this line with analytic weight formulations, and follow-up work revisits weight construction through staged refinement [43] or learned priors [299].

Mesh rendering instead optimizes a sampled field, extracts an explicit mesh with a differentiable iso-surfacing layer, and supervises that mesh through differentiable rasterization. Nvdiffrec [176] instantiates this pipeline with DMTet [223] and jointly optimizes geometry, materials, and lighting. FlexiCubes [224] extends DMC on regular cube grids with flexible vertex positioning and per-cell topology control, whereas TetWeave [11] replaces the predefined tetrahedral grid of DMTet with on-the-fly Delaunay tetrahedralization for adaptive resolution.

Comparison of rendering strategies. Surface rendering is geometrically direct, but each ray supervises only one surface point, making optimization sensitive to self-occlusion, thin structures, and abrupt depth changes. Volume rendering distributes supervision over all samples along a ray, greatly improving optimization robustness, but requires many field evaluations and carefully designed field-to-weight mappings. Mesh rendering optimizes geometry through an extracted mesh

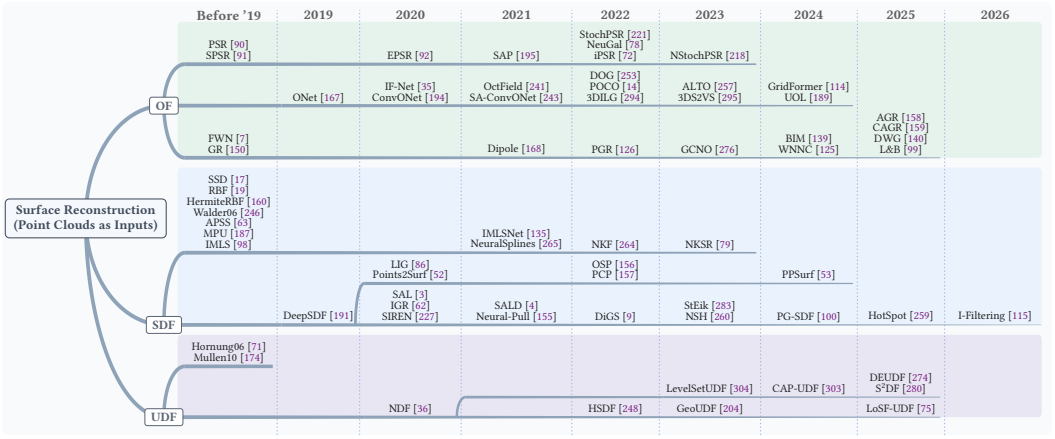


Fig. 9. Evolution of implicit surface reconstructions (3D point clouds as inputs).

and differentiable rasterization, making it efficient and directly amenable to joint material and lighting estimation, but its fidelity is bounded by the underlying grid resolution and iso-surfacing quality.

Hybrid representations. Hybrid methods co-optimize the implicit field with an explicit proxy that provides dense geometric cues or efficient rendering, while the implicit field supplies a smooth, extractable surface. Sur^{2}f [80] pairs an SDF with a deformable surrogate mesh, synchronizing them through a shared neural shader and using the mesh for surface-guided volume sampling. More recently, 3D Gaussians [93] have become a popular proxy owing to their fast differentiable rasterization. Several methods couple 3D Gaussians with a neural SDF, using the Gaussians to provide geometric priors or rendering supervision and the SDF to regularize the underlying geometry [23, 153, 291]; GaussianUDF [116] extends this hybrid paradigm to open surfaces by pairing Gaussians with a UDF.

4.2 3D Point Clouds as Inputs

Point clouds are a fundamental 3D sensing modality, but their discreteness, lack of connectivity, and missing or inconsistent orientation make surface reconstruction ill-posed. Implicit fields are well suited to this setting because point samples naturally provide local constraints on the field function f, e.g., $f(\mathbf{p}) = 0$ for surface samples and $\nabla f(\mathbf{p}) = \mathbf{n}$ for oriented points.

We organize these methods along the two families introduced above: *classical* models define f through explicit mathematical or physics-inspired formulations, often gaining stronger built-in geometric regularity; *neural* models parameterize f with coordinate networks, gaining greater modeling flexibility and the ability to incorporate learned priors. Fig. 9 summarizes the evolution of these methods.

4.2.1 Classical Implicit Field Reconstruction. Classical implicit methods define f through prescribed mathematical or physics-inspired structures rather than a learned coordinate network. These structures give explicit control over regularity and often yield robust, watertight reconstructions, but many formulations are well-posed only when globally consistent normal orientation is available.

Oriented-point methods. When globally consistent normals are available, reconstruction is much better conditioned: normals provide both local surface orientation and the sign cues needed by many classical formulations. The main distinction is then how the oriented input determines

f: as the solution of a global variational problem, by interpolation or approximation within a prescribed function family, or through an integral formula induced by the input.

Variational PDE methods. These methods define f implicitly as the minimizer of a global variational objective. Poisson Surface Reconstruction (PSR) [90] is the canonical example: it interprets the oriented normals as samples of the gradient of an indicator function, interpolates them into a dense vector field $\mathbf{V}$, and solves the corresponding Poisson problem to recover an indicator-like field whose gradient best approximates $\mathbf{V}$. The global solve produces an inherently watertight surface, but constraining only ∇f leaves the absolute field value unanchored, requiring a post-hoc iso-value choice. Screened PSR [91] and SSD [17] both add position fidelity, the former within the original indicator-function framework and the latter by directly targeting a smooth approximate SDF. Follow-up work extends the same variational route with envelope constraints from visual or depth hulls [92], uncertainty estimation through stochastic reformulations [221, 218], or learned spatially adaptive basis functions while still solving for the coefficients through a linear system [78].

Interpolation and approximation methods. These methods prescribe an explicit functional form for f and determine it from the samples, either by global interpolation through the data or by local approximation near each query point.

Global interpolation methods construct f as a weighted sum of basis kernels centered at the samples and solve for the coefficients; the choice of kernel then governs both the smoothness of the reconstruction and the computational cost. RBF [19] places a globally supported polyharmonic kernel at each sample and fits the field by enforcing value constraints on and off the surface, yielding a smooth interpolant. HRBF [160] replaces the off-surface constraints with direct Hermite conditions on both values and gradients. Neural Splines [265] substitutes the hand-designed kernel with the arc-cosine kernel of infinitely wide shallow ReLU networks, drawing the prior from the corresponding neural RKHS. Walder et al. [246] switch to compactly supported multi-scale bases with a global regularizer, yielding sparse systems that scale to larger inputs.

Local approximation methods instead evaluate f at each query point from fits to nearby samples. IMLS [222, 98] computes a locality-weighted average of signed distances to tangent planes at nearby oriented samples. APSS [63] replaces tangent planes with algebraic spheres for better stability under high curvature. MPU [187] blends local quadratic approximants on an adaptive octree through partition-of-unity weights.

Recent learned variants preserve this explicit evaluation structure while learning components of the interpolation or approximation family. NKF [264] learns a global kernel prior and reconstructs by kernel ridge regression, while NKSR [79] replaces global kernels with compactly supported ones and adds gradient fitting, scaling to large scenes. IMLSNet [135] learns the local MLS parameterization, predicting the octree scaffold and primitives from which the field is evaluated in closed form.

Integral methods. These methods define f directly by integrating contributions from the oriented input rather than solving for unknown coefficients. GWN [85] extends the classical winding number to oriented triangle soups, yielding an inside–outside field that degrades gracefully under gaps and self-intersections. FWN [7] transfers this idea to oriented point clouds via a hierarchical double-layer potential approximation: $f(\mathbf{q}) = \sum_{i=1}^{m} a_i \, (\mathbf{p}_i - \mathbf{q}) \cdot \hat{\mathbf{n}}_i / 4\pi \, \|\mathbf{p}_i - \mathbf{q}\|^3$, where $\mathbf{p}_i$, $\hat{\mathbf{n}}_i$, and a_i are the position, outward unit normal, and area weight of each oriented sample. GR [150] derives a related field from a modified Gauss formula, but regularizes the near-surface singularity so that the field behaves like an indicator far from the surface and like an SDF near it.

Unoriented-point methods. Without reliable normals, the main difficulty is recovering consistent sign information. Existing methods either orient the input first or estimate orientation jointly with reconstruction; early work also explored postponing signing to a post-processing step [71, 174].

Orientation as preprocessing. The standard pipeline follows Hoppe et al. [70]: estimate unoriented normals by local PCA, then propagate a consistent orientation along an MST before applying any oriented solver. PCPNet [64] improves the local normal estimates with a learned patch encoder, while Dipole [168] replaces MST traversal with dipole-based long-range propagation. More recent work treats orientation as an inverse problem on the winding-number field, optimizing candidate normals until the induced field satisfies desired structural properties. GCNO [276], BIM [139], and WNNC [125] respectively encourage approximate binarity, low Dirichlet energy, or gradient-normal consistency of the induced field.

Joint orientation and reconstruction. These methods fold orientation into the reconstruction objective itself, so that the field and the normals are coupled and solving for one constrains the other. iPSR [72] extends Screened PSR by alternating its global solve with normal re-estimation from the current surface. PGR [126] extends GR by treating the per-point surface elements as unknowns and solving for them jointly with the indicator field in one linear system; AGR [158] and CAGR [159] augment PGR with anisotropic fundamental solutions that reduce underdeterminedness, especially on thin structures. DWG [140] alternates between recomputing a screened winding field and updating normals from its level-set gradients until both stabilize. VIPSS [82] works in the same kernel function space as the classical oriented fitting methods above but treats the normals as free variables, finding the smoothest interpolant through the input points by minimizing a global biharmonic energy that couples all normals into a consistent configuration. NN-VIPSS [269] localizes VIPSS to natural neighborhoods, scaling to much larger point sets.

4.2.2 Neural Implicit Field Reconstruction. Neural implicit field methods instead represent f as a coordinate network. We group them by how this network is obtained: *learning-based prediction*, *per-instance optimization*, and *hybrid methods*.

Learning-based prediction. These methods train an encoder together with the coordinate decoder on shape datasets, so reconstruction reduces to a single forward pass at test time. They differ mainly in how the input conditions the decoder.

Spatial feature conditioning. The simplest version conditions the decoder on a single global latent code, as in ONet [167], but the fixed-size bottleneck limits fine detail. Later work therefore conditions each query on spatially organized features. IF-Net [35] voxelizes the input, learns multi-scale 3D features, and interpolates them at query points before decoding. ConvONet [194] instead projects point features onto canonical planes or volumetric grids and refines them convolutionally, yielding a scalable framework for large scenes. Subsequent work pushes the same idea toward adaptive octree features [241, 253], point-native latents [14], token-based spatial latents for transformer-style decoding [294, 295], and explicit point-grid interaction schemes [257, 114].

A parallel line extends IF-Net's multi-scale encoding to non-watertight geometry. NDF [36] regresses unsigned distances and extracts surfaces by gradient-based projection. To enable direct mesh extraction on open surfaces, HSDF [248] additionally predicts a sign field, while GIFS [290] predicts a segment field with an auxiliary UDF.

Query-centric feature conditioning. Instead of sampling from a pre-built spatial structure, these methods construct features from the local neighborhood of each query point, naturally adapting to varying density. Points2Surf [52] decomposes SDF prediction into a local patch-based distance regressor and a global sign classifier; PPSurf [53] extends it by fusing global context with query-local priors through attention. For open surfaces, GeoUDF [204] aggregates unsigned distances to tangent planes induced by local quadratic patches, while LoSF-UDF [75] learns a local UDF prior from synthetically generated analytic patches. SALS [206] applies the same local-query conditioning idea to a segment field, enabling direct mesh extraction on surfaces with arbitrary topology.

Per-instance optimization. These methods fit a neural field directly to a single point cloud, without dataset-level training. The main distinction is where the fitting signal is imposed: directly on the field, or indirectly through projections onto the current surface.

Fitting via direct field supervision. Methods in this family supervise the neural field f by placing losses directly on its output values and/or their spatial derivatives at sampled points.

IGR [62] establishes the foundational paradigm: on-surface zero-value constraints anchor the field to observations, while an Eikonal regularizer $\|\nabla f\| \approx 1$ throughout the domain encourages global SDF behavior. SIREN [227] replaces the ReLU MLP with sinusoidal activations whose derivatives at any order remain non-degenerate, enabling exploitation of higher-order differential properties. Building on this route, DiGS [9], StEik [283], and NSH [261] all introduce second-order geometric regularization with different goals: DiGS penalizes the Laplacian to stabilize fitting on unoriented inputs; StEik replaces the Laplacian with a normal-direction second-order term to avoid over-smoothing; NSH enforces a near-surface singular-Hessian prior to suppress ghost geometry. HotSpot [259] augments the objective with a screened Poisson term, turning the necessary Eikonal constraint into an asymptotically sufficient one.

An orthogonal sign-agnostic line fits SDFs from unsigned data. SAL [3] regresses $|f|$ to unsigned distance values and relies on sphere initialization to resolve the sign ambiguity, while SALD [4] further incorporates near-surface derivative information of the unsigned distance function to improve sample efficiency and sharp-feature recovery.

Beyond SDFs, DEUDF [274] targets high-fidelity UDF fitting with normal-alignment cues and adaptive near-surface regularization, while S^2DF [280] replaces UDF with a differentiable scaled-squared distance representation paired with a Monge–Ampère-type regularizer.

Fitting via projection-based supervision. Neural-Pull [155] supervises the field indirectly through its induced surface projection: it projects each query onto the zero level set along the predicted gradient by the predicted distance, and minimizes the gap between the projected point and its nearest neighbor in the input. This pulling loss implicitly couples field values with gradient directions without an explicit Eikonal term. I-Filtering [115] extends this with an implicit bilateral filter that extends to non-zero level sets, enforcing inter-level-set geometric consistency to regularize the zero level set while preserving high-frequency details. Follow-up works adapt the pulling mechanism to UDFs [303, 304], extending this route to open and non-manifold surfaces.

Bridging the direct and projection-based routes, PG-SDF [100] combines losses from both and addresses initialization sensitivity through point-guided homotopy optimization.

Hybrid methods. Hybrid methods combine learned priors with per-instance optimization to balance generalization and instance-specific fidelity. DeepSDF [191] trains a shared SDF decoder via an auto-decoder framework with per-shape latent codes; at test time a new code is fitted to observations while staying close to the learned shape manifold. LIG [86] extends this to large scenes via overlapping local latent codes. SA-ConvONet [243] initializes from a ConvONet prediction and applies sign-agnostic test-time optimization to resolve sign conflicts. OSP [156] and PCP [157] train auxiliary prior networks and incorporate them as differentiable supervision during per-instance SDF optimization, improving robustness under severe sparsity and noise.

5 3D GEOMETRY GENERATION

The reconstruction methods of Sec. 4 seek a single implicit field that best explains the observations. This deterministic formulation works well under strong conditioning but tends to average over plausible modes as constraints weaken, producing overly smooth surfaces. Geometry generation addresses this limitation by modeling a distribution over implicit fields, enabling diverse, detail-rich samples even from weak conditioning such as a single image or a text prompt. The scarcity of

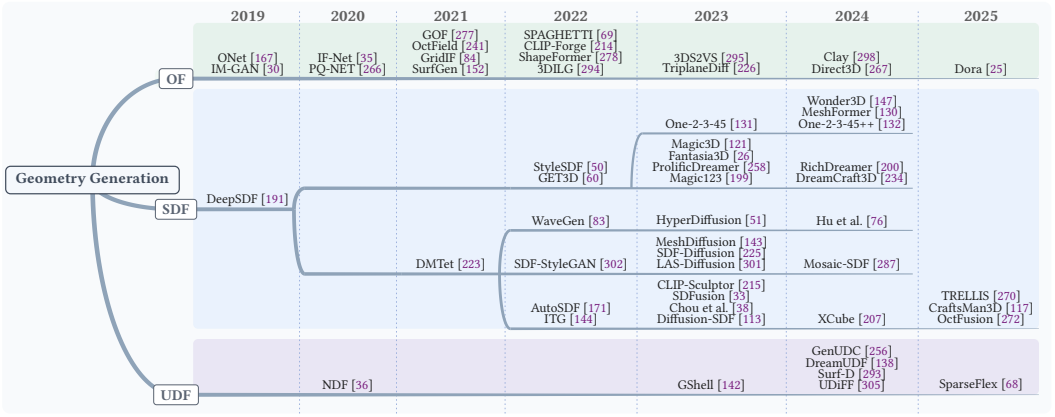


Fig. 10. Evolution of implicit geometry generation.

large-scale 3D shape collections and the high dimensionality of implicit fields have led to two major lines of work.

2D-lifted methods bypass the data bottleneck by constraining 3D generation through supervision in 2D image space. **Native 3D methods** learn the shape distribution directly from 3D data, tackling dimensionality through compact latent representations. Fig. 10 summarizes the evolution of these methods.

5.1 2D-Lifted Generation

2D image priors support 3D generation either by directly supervising a 3D representation in image space or by first generating intermediate views that are then used for 3D reconstruction. We review three regimes: *feed-forward generation with adversarial supervision*, *per-instance optimization with diffusion priors*, and *feed-forward reconstruction from diffusion-generated views*.

5.1.1 Feed-forward Generation with Adversarial Supervision. The earliest 2D-lifted methods train a shared generator under adversarial image supervision: the generator produces an implicit field, a differentiable renderer maps it to images, and a discriminator scores the renderings. Representative works progressively shift the underlying representation from NeRF-based OFs [277] to SDFs with volume-rendered features [50], and further to DMTet-extracted meshes supervised through rasterization [60]. These models are efficient at inference, but adversarial training is prone to mode collapse and requires category-specific image collections, restricting open-domain generation.

5.1.2 Per-instance Optimization with Diffusion Priors. These methods fit a 3D representation to each input by distilling frozen diffusion priors at test time. DreamFusion [198] introduced this regime with Score Distillation Sampling (SDS) on a NeRF, enabling open-domain 3D generation from text without any 3D training data. Follow-up work shifts the underlying representation toward surfaces: Magic3D [121] and Fantasia3D [26] adopt DMTet for explicit mesh extraction, while DreamUDF [138] extends the regime to open surfaces via UDF. Concurrent efforts improve distillation quality through variational formulations [258] or mitigate the Janus problem with 3D-aware diffusion priors [133, 199, 234, 200]. Despite broad applicability, each result still requires hundreds to thousands of optimization steps.

5.1.3 Feed-forward Reconstruction from Diffusion-Generated Views. Rather than distilling diffusion priors directly into a 3D field, this regime first synthesizes multi-view intermediates and then

reconstructs 3D from them. Building on Zero-1-to-3 [133], subsequent methods refine this generate-then-reconstruct paradigm by improving multi-view synthesis, downstream 3D reconstruction, or both [131, 147, 130, 132].

5.2 Native 3D Generation

Native 3D generation learns the shape distribution directly from 3D data itself, rather than constraining 3D generation through 2D image priors or rendered image supervision. Its main challenge is scalability: the parameter space of an implicit representation is high-dimensional, and dense volumetric discretizations grow cubically with resolution. Existing methods mainly differ in where the generative model operates. *Direct generation* works in the original field space, whereas *latent generation* first compresses shapes into a lower-dimensional representation and models the distribution there.

5.2.1 Direct Generation. Direct generation applies the generative model directly to the original 3D representation without first compressing shapes into a latent code.

Adversarial methods. These methods differ mainly in where the discriminator operates. DMTet [223] extracts an explicit mesh from an SDF on a tetrahedral grid and supervises the mesh surface adversarially. SurfGen [152] projects the zero iso-surface onto the unit sphere and applies a spherical CNN discriminator. SDF-StyleGAN [302] stays entirely in the implicit domain, applying global and local discriminators to SDF values and gradients.

Diffusion and flow-matching methods. These methods replace adversarial supervision with diffusion or flow matching. One route operates directly on SDF volumes. SDF-Diffusion [225] uses a cascaded pipeline that first generates a low-resolution field and then super-resolves it patch by patch. LAS-Diffusion [301] adopts a two-stage occupancy-then-SDF pipeline with local attention for sketch-conditioned generation. Mosaic-SDF [287] departs from dense grids, representing each shape with a sparse set of local SDF patches near the boundary and modeling them with a transformer-based flow-matching network. Another route couples diffusion with structured mesh-related representations. MeshDiffusion [143] generates watertight meshes by denoising the SDF values and vertex deformations of a DMTet grid. GShell [142] further handles open shapes by defining a manifold SDF on a watertight shell enclosing the target.

5.2.2 Latent Generation. Latent generation shifts distribution modeling from the full 3D field to a compact latent space. Shape codes are obtained either by *amortized encoding*, which learns a shared encoder–decoder over a shape collection, or by *per-shape derivation*, which obtains each code independently.

Amortized encoding. Early methods such as ONet [167] and OctField [241] adopt a single-stage VAE framework, regularizing the latent space toward a unit Gaussian for direct sampling. Yet a unimodal prior poorly captures multi-modal shape distributions, motivating two-stage pipelines that first train an autoencoder for reconstruction and then learn the latent distribution with a more expressive model. Early two-stage methods explore increasingly structured codes, from global vectors [30] to spatially localized grids [84] and part-level assemblies [69, 266], with text-conditioned variants leveraging CLIP features [214, 144].

With diffusion replacing these earlier priors, the latent representation has continued to evolve. Early methods compress voxelized SDFs into compact spatial latent grids [113, 33] or into global latent vectors via modulation-based SDF-VAEs [38]. Subsequent work replaces the dense grid either with unstructured latent vector sets [295], a design later scaled and refined in [298, 25, 117], or with sparse spatial structures such as octrees and sparse voxel grids [272, 207, 270], with Sparse-Flex [68] further supporting open surfaces through sparse FlexiCubes. Other directions include triplane-factorized latents [267] and extending the underlying field to UDFs for arbitrary-topology

generation [293]. GenUDC [256] applies latent diffusion to the face-connectivity component of an Unsigned Dual Contouring representation, then predicts the corresponding vertex positions. A parallel discrete branch quantizes latents into codebook tokens for autoregressive generation, exploring various spatial tokenization strategies [278, 171, 294], with CLIP-Sculptor [215] extending this to zero-shot text-to-shape synthesis.

Per-shape derivation. Unlike amortized encoding, the following methods derive each shape's latent independently. One line decomposes TSDF grids into wavelet coefficients and applies diffusion in the wavelet domain [83, 76], with UDiFF [305] extending this to UDFs for open-surface generation. Triplane Diffusion [226] fits each shape to tri-plane feature maps and trains a 2D diffusion model on those maps. HyperDiffusion [51] takes a more direct route, diffusing over the full weights of per-shape overfit MLPs.

6 BENCHMARKING IMPLICIT REPRESENTATION-BASED 3D SURFACE RECONSTRUCTION FROM 3D POINT CLOUDS

Point cloud surface reconstruction is the longest-standing application of implicit fields, studied through both classical and neural formulations as reviewed in the preceding sections. Recent benchmarking efforts [81] have advanced standardized evaluation, yet their focus on watertight geometry leaves non-watertight surfaces with open boundaries and non-manifold edges underexplored. To fill this gap, we benchmark 19 representative methods across four complementary scenarios covering both object-level geometry (watertight and non-watertight) and scene-level data (synthetic and real-scanned) with varying point cloud densities and additional noise levels for object-level shapes.

6.1 Experimental Setup

We evaluate 19 methods spanning classical fields, learning-based neural prediction, and per-instance neural optimization across four scenarios: *watertight objects*, *non-watertight objects* containing open boundaries and non-manifold topology, *synthetic scenes* with clean ground truth, and *real-scanned scenes* with complex topologies and noise.

Methods are grouped by input configuration (oriented versus unoriented), with some evaluated under both settings; unoriented methods are further divided into closed-surface and open-surface categories, of which only the latter is applicable to non-watertight and real-scanned data.

For each object, we generate point clouds at three densities (3K, 30K, and 100K points, termed Sparse, Clean, and Dense) and two noisy variants by adding isotropic Gaussian noise with standard deviation $\sigma \in \{0.002\ell, 0.005\ell\}$ to the 30K configuration, where ℓ is the bounding-box diagonal. Scene point clouds are sampled at 500 and 1000 points per m^2, yielding 330 evaluation instances in total. All methods are compared using Chamfer Distance (CD), F-Score (F1), Normal Consistency (NC), and Hausdorff Distance (HD). We additionally report point-cloud-to-field time (T_{field}), field-to-mesh time (T_{extract}), and peak GPU memory usage (VRAM). Full details of the evaluation scenarios, evaluated methods, and evaluation protocol are provided in Sections B.1–B.3 of the supplementary material.

6.2 Results and Analysis

Full quantitative results, including reconstruction accuracy, runtime, and peak GPU memory usage, are provided in Section B.4 of the supplementary material.

Oriented normals remain a powerful but no longer indispensable prior. Under clean, dense conditions the best unoriented methods match or even surpass neural oriented methods on watertight objects (Supplementary Table 3), yet classical field methods retain a unique robustness advantage under noise and at scene scale (Supplementary Table 5): SPSR leads across nearly all metrics in these regimes, as its prescribed functional structure and global variational solve

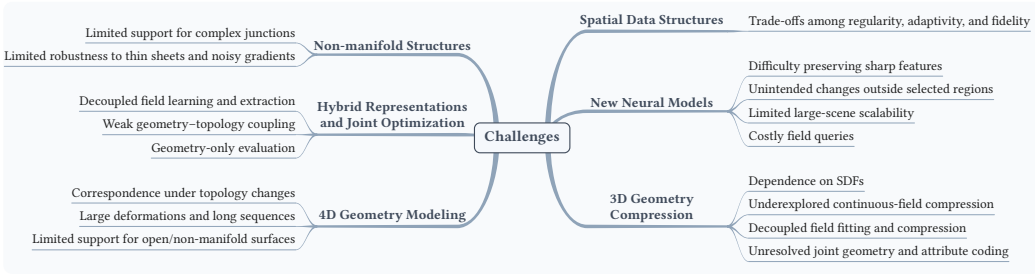


Fig. 11. Overview of challenges in implicit 3D geometry.

consider all points simultaneously, making it inherently resilient to local perturbations in a way that pointwise neural fitting cannot easily replicate.

Open-surface methods match closed-surface methods on watertight geometry while naturally extending to open topology. Within the unoriented group on watertight objects (Supplementary Table 3), open-surface methods such as S^2DF and GeoUDF achieve comparable accuracy to closed-surface methods under most conditions, while additionally handling non-watertight topology where closed-surface formulations are structurally inapplicable.

Data sufficiency governs the learning-versus-optimization trade-off. Under sparse input (Supplementary Tables 3 and 4), per-instance optimization methods such as S^2DF degrade sharply as too few constraints remain to resolve surface geometry, while learning-based methods such as GeoUDF maintain reasonable accuracy thanks to pretrained shape priors. The advantage reverses at higher densities, where per-instance optimization exploits fine-grained local detail more fully than fixed feed-forward predictors.

Additional results on mesh topology and triangle quality for watertight objects are provided in Section B.5 of the supplementary material.

7 CHALLENGES, FUTURE DIRECTIONS, AND OTHER APPLICATIONS

Taken together, the preceding sections show that implicit geometry is best understood not as a collection of isolated field definitions, but as a coupled pipeline in which representation, extraction, supervision, and application constraints continuously shape one another. Our survey of occupancy, signed distance, unsigned distance, and related implicit fields clarified the distinct trade-offs each formulation makes in regularity, topology, geometric queries, and downstream compatibility. These observations suggest that the central problems ahead are no longer simply which field to use, but how to design implicit representations and extraction pipelines that are simultaneously more robust, topology-aware, data-efficient, scalable, and controllable. With this perspective in place, we now turn to the major challenges and future research directions in implicit representation and extraction, as well as in further geometry-processing tasks. Fig. 11 summarizes the main challenges across these topics. The section concludes with an overview of applications in other domains.

7.1 Non-manifold Structures

Many geometric objects of practical interest are intrinsically non-manifold: material interfaces may meet along shared curves or higher-order junctions, medial axes are inherently sheet-based structures with merges and terminations, and man-made objects often contain thin plates, nested shells, internal partitions, and multi-layer components that create non-manifold configurations.

In these settings, a globally consistent inside/outside labeling becomes ambiguous or ill-posed, motivating unsigned and medial implicit representations. The main challenge, however, lies not only in representation but also in extraction: recovering consistent sheet connectivity near intersections,

mergers, and branch junctions. Optimization-based methods such as DCUDF [73] and DCUDF2 [28] can produce a double-layered mesh around the surface, including around non-manifold junctions. However, converting this two-layer membrane into a single non-manifold mesh remains challenging, because front/back separation breaks down at Y-shaped and higher-order junctions, as noted in [103].

Recent efforts have begun to address this challenge from different perspectives. DualMesh-UDF [296] can recover non-manifold configurations, but its cell-wise connectivity remains fragile in ambiguous cells, leading to holes or other topological artifacts. MIND [29] supports non-manifold extraction through multi-label interfaces, but cannot handle non-orientable surfaces, and its cube-based discretization limits the number of locally separable sheets. VoroUDF [251] offers greater flexibility through Voronoi-based connectivity, accommodating more complex non-manifold configurations. DCx [6] adapts Dual Contouring to UDFs through optimization-guided active-cube detection and a voxel-to-mesh lookup table, supporting non-manifold connectivity within expanded cubes.

Despite recent progress in extracting iso-surfaces with non-manifold structures, two important directions remain open. First, current extractors still provide only limited support for arbitrary junction complexity. Grid-based methods encode a restricted family of local configurations, whereas Voronoi-based methods improve flexibility but still lack formal topological guarantees. Developing topology-complete extractors that can robustly recover high-valence junctions, non-manifold points, and eventually non-orientable structures remains an important open problem. Second, robustness to thin layers, closely spaced sheets, and noisy gradients remains limited. This points to the need for adaptive resolution, uncertainty-aware extraction, and explicit topological validation or correction.

7.2 Hybrid Representations and Joint Optimization

Hybrid representations combine complementary field types so that geometric structures that are difficult to recover with any single representation, especially open surfaces, thin sheets, and medial axes, can still be extracted through standard iso-surface pipelines.

A common principle is to decouple metric proximity from topological validity: one component models distance in a way that remains meaningful across the domain, while another determines where sign, phase, or surface existence should be trusted. This idea appears in several forms for surface reconstruction. Points2Surf [52] is an early precursor to hybrid metric–sign decomposition. It factorizes the SDF into an absolute-distance term and a sign component, predicts the absolute distance from local point-cloud patches, estimates the sign from coarse global context, and reconstructs the final SDF as their product before applying Marching Cubes. Hybrid Sign and Distance Field (HSDF) [248] extends this decomposition to arbitrary-topology surface reconstruction. It learns a UDF together with an auxiliary sign field, uses the UDF to localize candidate surface points, and then applies an adaptive masked MC procedure that instantiates a mesh only in surface-supported regions. In this way, HSDF combines UDF-style topological flexibility with SDF-style field-to-mesh conversion, enabling both open and closed surfaces to be reconstructed in a unified framework. SSDF [39] combines a generalized winding-number signal with an unsigned distance field, yielding a semi-signed distance representation that retains proximity information while representing holes and open boundaries implicitly. Three-pole Signed Distance Function (3PSDF) [27] takes a different route by extending the conventional inside/outside signed distance formulation with a third, directionless NULL sign. By assigning null values to regions that contain no target surface, it suppresses unwanted decision boundaries and prevents the implicit field from closing open structures. To avoid the gradient difficulties caused by null-valued targets, 3PSDF formulates learning as a three-way classification problem over {inside, outside, null}, while retaining compatibility with

Marching Cubes for mesh extraction. Neural rendering with Arbitrary Topologies (NeAT) [166] brings the same idea into the multi-view neural rendering setting. It represents geometry with an SDF augmented by a differentiable validity branch that predicts the probability that a surface should exist at a queried 3D point. During rendering, low-validity samples are suppressed, and at extraction time low-validity regions are masked out before Marching Cubes by setting their SDF values to `NaN`. This allows NeAT to reconstruct open as well as watertight surfaces while preserving the rendering and meshing advantages of signed implicit fields. Metric–Phase Fields (MPF) [102] decouple geometry and topology more explicitly by learning an unsigned metric field together with a smooth phase field; the two are fused into a signed implicit function through a gated residual formulation, allowing the representation to behave like an SDF where sign is meaningful while remaining UDF-like in sign-ambiguous thin structures and boundary regions. Unlike Points2Surf [52] and HSDF [248], which decode both sign and distance from a shared feature space, MPF learns them separately and introduces a soft constraint to encourage their alignment along the gradient direction. Such alignment improves numerical stability near the surface, where small inconsistencies can otherwise lead to artifacts.

Hybrid representations are also closely related to joint optimization. For instance, GIFS [290] learns a segment field jointly with a UDF under a shared backbone. The segment field classifies whether two query points are separated by a surface, providing the region-partitioning cues that a standalone UDF lacks for Marching Cubes extraction; the UDF in turn supplies metric proximity for localizing intersection points. In room-level reconstruction, Occ-SDF Hybrid [154] and H2O-SDF [192] each extend a volume-rendered SDF framework with a second field to recover thin indoor structures. The former jointly trains an OF whose per-point occupancy is unaffected by other objects along the ray, compensating for the density distortion that SDF incurs on multi-object rays; the latter introduces an Object Surface Field, a learned surface-probability field whose cross-guidance loss creates non-vanishing gradients with respect to SDF values, enabling convergence at thin structures that would otherwise be lost to vanishing gradients. Q-MDF [103] jointly learns an SDF and a medial field [201], and defines a quasi-medial distance whose zero level set recovers the medial axis.

Across these hybrid representations and joint optimization methods, the key advantage is that iso-surface extraction is applied only where the auxiliary sign, validity, or phase information is meaningful, which suppresses the unwanted closures and topological distortions that arise when a single signed field is forced to explain open or sign-ambiguous geometry.

Several future directions appear particularly important for hybrid representations. First, representation and extraction remain largely decoupled: the field is learned first, and topology is recovered afterward using Marching Cubes variants or post-processing. A promising direction is therefore to co-design hybrid fields with topology-aware extraction structures, such as material-interface partitions [29], adaptive cell complexes [6], or Voronoi-based discretizations [251], so that topology is enforced during learning rather than repaired afterward.

Second, the coupling between geometric and topological components is often still weak. Most existing methods combine metric and sign/validity cues in a loosely connected manner, which can lead to inconsistencies near boundaries, thin structures, and junctions. This suggests the need for tighter analytical and optimization-level coupling, including uncertainty-aware phase prediction, adaptive transition sharpness, and explicit consistency constraints near boundaries and high-order junctions.

Third, since hybrid representations are typically designed to address challenging scenarios where conventional implicit functions fail, evaluation protocols should move beyond purely geometric errors. Metrics such as Chamfer distance or Hausdorff distance are insufficient to capture the intended improvements. More informative evaluation should explicitly measure boundary

fidelity, thin-sheet separation, phase/sign consistency, and junction correctness, particularly for non-manifold configurations.

7.3 Spatial Data Structures

Spatial data structures do not merely accelerate implicit methods; they also define the neighborhoods over which fields are optimized, regularized, and extracted. In practice, different structures induce different notions of locality and therefore different trade-offs between regularity, adaptivity, and geometric fidelity.

Octrees remain the most commonly used spatial scaffold for implicit reconstruction. In the Poisson Surface Reconstruction family, PSR [90] uses an adaptive octree to represent the implicit function and solve the Poisson system only where detail is needed; SPSR [91] retains the same hierarchy while adding interpolation constraints; and iPSR [72] extends the framework to unoriented points by iteratively updating normals from the reconstructed surface. The same logic persists in recent learned-UDF meshing: DualMesh-UDF [296] uses an adaptive octree to localize tangent-plane estimation and surface-point placement.

Adaptive grids and sparse voxel structures pursue a similar principle, but with a more explicit notion of occupancy. Rather than storing features over a dense lattice, methods such as NSVF [129] organize voxel-bounded implicit fields in a set of sparse voxels, allowing computation to skip empty space and focus on occupied regions. Voxurf [268] follows the same philosophy for neural surface reconstruction: it combines an explicit voxel-based SDF representation with a two-stage coarse-to-fine optimization procedure and hierarchical geometry features that encourage information sharing across voxels, thereby improving both efficiency and surface fidelity. Taken together, sparse voxels and adaptive grids can be understood as generalizations of the octree philosophy: they couple implicit functions to discretizations that place representational capacity where geometry is present, intricate, or topologically ambiguous.

Voronoi-based structures offer a complementary, sample-aware alternative to voxel hierarchies. In GCNO [276], Voronoi vertices and outside Voronoi poles are used to regularize the winding-number field and recover globally consistent normal orientations from raw point clouds; BIM [139] is a closely related follow-up in the same winding-number line that replaces this with a boundary-integration energy; and VAD [104] uses Voronoi-based geometric criteria to assign bidirectional normals before diffusing them into an approximate UDF gradient field. At the meshing stage, VoroUDF [251] goes further and replaces fixed grid cells with Voronoi cells, ultimately extracting connectivity from an approximate geodesic Voronoi diagram on the implicit zero set. Seen in this light, recent Voronoi-based neural/UDF methods reconnect with older Voronoi-interface work in multiphase implicit geometry, where Voronoi partitions were already used to track and mesh interconnected implicit interfaces.

Point-based structures are equally important, especially when the input is irregular, sparse, or thin. Point Set Surfaces [2] treats the point cloud itself as the primary geometric carrier and defines a smooth surface via local moving-least-squares projections. VIPSS [82] reformulates point-based surface reconstruction as a variational implicit problem on unoriented points, showing that the field can be defined directly from points rather than from a volumetric grid. NN-VIPSS [269] makes this idea substantially more scalable by exploiting the locality of natural neighborhoods, effectively turning a global variational formulation into a locally supported one without sacrificing robustness on sparse and non-uniform inputs. This point- and natural-neighbor-based perspective is particularly attractive for thin structures, where fixed voxel neighborhoods can be too rigid and geometry-aware local support is often more reliable.

Finally, multiresolution hash tables have become a dominant learned spatial structure for neural implicit fields. Instant-NGP [175] introduces the multiresolution hash table as a compact trainable

encoding for coordinate networks. Neuralangelo [119] shows how multi-resolution 3D hash grids, combined with coarse-to-fine optimization, can drive high-fidelity neural surface reconstruction. Although these encodings are not meshing structures in the classical sense, they play a closely related role: they determine where representational capacity is concentrated in space, and thus strongly influence both optimization and the fidelity of the learned implicit function.

7.4 New Neural Models

Recent work has increasingly moved beyond the standard formulation of neural implicit fields as a single smooth coordinate-based MLP representing an occupancy function, SDF, or UDF. These new neural models can be understood from two perspectives: representation power and downstream usability. The former concerns how network architectures can better encode geometric, topological, or analytic structures that are difficult for generic coordinate MLPs to learn, while the latter concerns how neural implicit fields support practical downstream use beyond pure representation quality.

From the representation perspective, a central challenge is to model smooth surface patches while preserving the sharp edges and corners where they meet. Specialized fitting objectives and geometric priors improve feature reconstruction [115, 48, 254]. However, MLPs with smooth activations cannot represent gradient jumps, while nonsmooth activations alone do not ensure that these jumps align with geometric features [185]. To encode sharpness explicitly, NH-Rep [66] combines smooth implicit fields for a given CAD patch decomposition through Boolean operations, forming sharp features at patch intersections. Patch-Grid [124] localizes these operations to limit interference between unrelated patches.

SharpNet [185] instead introduces nonsmoothness through an auxiliary network input, an idea also explored for function-value jumps in 2D neural fields [8, 128]. Its auxiliary field remains continuous but has derivative jumps across prescribed surfaces, allowing creases where these surfaces intersect the zero level set. However, these feature surfaces require initialization from meshes or point clouds; optimization can refine their geometry but cannot repair topological defects. This motivates learning feature geometry and topology jointly from raw data.

During extraction, mesh vertices and edges must also align with the encoded sharp features (Sec. 3). DC [88] uses local tangent-plane constraints, while NMC [32] and NDC [31] learn vertex placement from neighboring samples, relaxing the grid-edge restrictions of MC. FlexiCubes [224] optimizes vertex placement and triangulation jointly with the field. Marching Neurons [231] instead analytically extracts the zero set of piecewise-linear neural fields, preserving their encoded sharp features. Extending this fidelity to more general neural representations at practical computational cost remains an important direction.

Beyond representation power, downstream usability raises a different set of requirements. Three directions are particularly important: local controllability for direct modeling and editing, scalable large-scene representations, and efficient field queries for downstream geometric computation.

First, local controllability requires changing selected regions while preserving the remaining geometry, yet updating the shared weights of a global MLP can affect the surface throughout its domain. One route therefore keeps the decoder fixed and ties field evaluation to editable geometric controls. NeuMesh [279] decodes SDF values from features and distance estimates associated with nearby mesh vertices, whereas Neural Generalized Cylinder [306] defines a neural SDF in coordinates relative to control curves and their cross-sections. Moving these controls changes the inputs to queries that depend on them, deforming the corresponding implicit surface. Another route fits representation parameters to prescribed geometric edits. Given an edited surface, Patch-Grid [124] refits features of edited and affected neighboring patches while keeping the decoder and other patch features fixed. Berzins et al. [10] instead relate parameter changes to surface motion and solve for updates that approximate prescribed displacements, including zero motion in regions

to preserve. When editing rearranges or replaces parts, adapting their connections becomes an additional concern. SPAGHETTI [69] lets users transform or replace individually encoded parts, then decodes their updated configuration into a single occupancy field. NeuForm [123] spatially blends parameters and features from two models: one fitted to the original shape to retain part details, and one trained across shapes to adapt connections to the edited arrangement. Learned part divisions may nevertheless differ from the regions a user wishes to edit, while displacement fitting is limited by the deformations the model can express. Future work could explore controls that adapt to user-selected editing regions and translate geometric operations into predictable field updates while preserving untouched geometry. Extending these capabilities to interactive shape construction could further develop neural implicit representations into tools for direct geometric design.

Second, scalability requires locality at the level of scene organization. A single global MLP is often inadequate for large objects or scene-level geometry, because its capacity is shared across the entire domain and distant regions are coupled through the same parameters. Recent work therefore treats scalability as a problem of spatial decomposition: instead of representing the whole scene with one global field, the model is organized into local implicit functions with controlled support, resolution, and coupling. Regular or overlapping local grids [86, 21], adaptive octree decompositions [241], and graph-based compositions of overlapping local SDFs [284] all follow this principle in different forms. The key challenge is not merely to increase model capacity, but to allocate it locally and adaptively while maintaining consistency across neighboring fields, so that large scenes can be represented without sacrificing fine geometric detail.

Third, efficient field queries are essential for downstream geometric computation. Meshing, rendering, collision detection, simulation, and path planning require repeated evaluations of field values, gradients, normals, or intersections. Recent architectures suggest that efficiency can be built into the implicit representation itself rather than handled only by external acceleration structures. A representative line is T-MLP [282], which attaches output tails to intermediate layers so that valid predictions can be obtained at different network depths. SAND [281] extends this idea by coupling the tailed network with a spatial network-depth map, allowing each query to be evaluated only to the depth required by its spatial importance and local geometric complexity. This turns neural field evaluation into an adaptive process and naturally supports level-of-detail control. Future work may further explore learned depth prediction, training-time adaptive computation, and redundancy-aware pruning, enabling more controlled trade-offs among accuracy, runtime, and memory.

7.5 Other Applications

Beyond reconstruction and generation, implicit fields support geometry compression by encoding fields instead of mesh connectivity [240, 239, 205, 247], and 4D modeling by deforming reference fields [184, 242, 109, 18, 249, 16, 37] or learning spatiotemporal fields [186, 161, 213, 212]. They also support spatial computing [119, 286, 263, 232, 1, 46] and medical imaging [173, 101, 45] through surface reconstruction and synthesis, and robotics [190, 118] and physical simulation [172, 42, 89] through direct geometric queries. Further details on these applications, representative methods, and challenges in compression and 4D modeling are provided in Supplementary Section A.

8 CONCLUSION

In this paper, we have presented a comprehensive survey of implicit geometric representations, their surface extraction algorithms, and their downstream applications from a representation-centric perspective. We unified diverse implicit representations as a pair $\mathcal{F} = (f : \mathcal{Q} \to \mathcal{V}, C)$ of a field function and a geometry condition. Under this formulation, we reviewed occupancy, signed

distance, and unsigned distance fields, together with their vector, ray, segment, medial, and hybrid extensions. Organized along field types, we further surveyed iso-surface extraction algorithms, from classical Marching Cubes and Dual Contouring to recent UDF-tailored variants. Building on this backbone, we then revisited two representative applications, namely surface reconstruction and 3D geometry generation. Within each application, we traced the evolution of methods along the axis of implicit representations. To assess the practical behavior of existing methods, we contributed a benchmark on point cloud surface reconstruction, the longest-standing application of implicit fields, covering watertight objects, non-watertight objects, synthetic scenes, and real scans. Under a unified protocol, the benchmark evaluates 19 representative classical and neural field methods. Finally, we discussed the major limitations of current implicit geometry methods and outlined several promising directions for future research. We hope this survey provides a useful reference for understanding the trade-offs of implicit representations and for designing methods that better align representation choice with algorithmic requirements and application goals.

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Implicit Representations of 3D Geometry: A Comprehensive Survey, Benchmark, and Future Directions (*Supplementary Material*)

A OTHER APPLICATIONS

A.1 Implicit Representation-based 3D Geometry Compression

Implicit geometry compression converts a mesh into an implicit field, compresses the field, and reconstructs the surface via iso-surface extraction, bypassing the need to code irregular mesh connectivity directly.

On dense grids, Tang et al. [43, 42] partition the TSDF into local blocks for real-time streaming of dynamic performance sequences, progressing from a linear codec to an end-to-end auto-encoder with a learned entropy model and lossless sign coding to preserve topology. NeCGS [35] targets geometry-set compression: each mesh is converted into a volume combining TSDF with per-vertex displacement vectors, and a shared auto-decoder amortizes inter-shape redundancy. SIR-SNC [45] moves to the sparse setting, retaining field values only on near-surface grid vertices and compressing the resulting sparse tensor with a sparse-convolutional auto-encoder.

Several future directions can substantially advance this area. Current methods operate almost exclusively on (T)SDF over regular or sparse grids; extending implicit compression to unsigned or hybrid representations would enable the coding of open surfaces, thin sheets, and non-manifold geometry that SDFs cannot faithfully encode. Continuous implicit fields, with their resolution-independent expressiveness, are also a natural fit for compression. Investigating compact architectures specifically optimized for this purpose remains largely unexplored, particularly in offline scenarios where higher encoding cost is acceptable. More fundamentally, existing pipelines first fit the implicit field and then independently compress it, yet jointly optimizing the representation together with compression could yield substantially better rate–distortion trade-offs. Finally, practical assets rarely consist of geometry alone, and jointly compressing geometry with surface attributes such as color and material parameters within a unified implicit framework is an important open problem that could build on recent advances in neural texture compression [13].

A.2 Implicit Representation-based 4D Geometry Modeling

Beyond static reconstruction, implicit fields also support the reconstruction and modeling of time-varying geometry without prescribing mesh connectivity. Existing methods either deform a reference field or model the field directly over space and time.

Deformation-factorized fields keep one 3D field and transport it through time with an explicit motion model, which in turn supplies dense correspondence. Methods trained across sequences amortize this construction over a dataset. Occupancy Flow [28] pairs a reference occupancy field with a velocity field continuous in time, giving the surface at any instant but only through a

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sequential integration, which LPDC [44] removes by regressing continuous reference-to-frame displacements that decode all frames in parallel. Both anchor the reference in the first frame, where the object may already be in an arbitrary pose. CaDeX [20] instead learns the reference in a canonical space reached by bijective maps from every frame, so that partial observations can be merged there before the geometry is encoded; composing such maps is a homeomorphism, however, so the surface topology can never change. Observing that a single global latent code generalizes poorly across identities, Motion2VecSets [5] distributes shape and deformation over latent vector sets [55] and models their distribution with diffusion, treating reconstruction from sparse observations as one-to-many rather than deterministic. Per-instance methods apply the same decomposition to a single sequence, where no prior learned across sequences compensates for noise or missing regions. CanFields [46] follows the flow-based pipeline of Occupancy Flow on raw scans, adding a consolidator that denoises each frame and weights it by an estimated confidence, so that observation errors are not accumulated into the canonical shape. NDR [4] and 4DRegSDF [7] instead supervise the canonical SDF by volume rendering [47, 53]: the former reaches it from RGB-D through an invertible deformation network, cycle-consistent by construction and made topology-variant by an auxiliary coordinate; the latter works from RGB alone, where deformation is otherwise constrained only by appearance, and regularizes it with a local-rigidity prior derived from the tangent planes and curvature of the moving zero level set.

Spatiotemporal fields instead admit time into the query domain, $f\colon \mathbb{R}^3 \times \mathbb{R} \to \mathbb{R}$, so that each instant is a slice of one field rather than a deformation of a reference. Variation of the field over time relates to surface motion through the level-set equation $\partial_t f + \mathbf{v} \cdot \nabla f = 0$, which fixes the normal component of the motion and leaves the tangential one free. NISE [29] instantiates this formulation, but evolves the surface under a prescribed velocity field rather than one inferred from observations. Neural SDF Flow [25] learns the temporal derivative of the SDF from multi-view video under volume rendering [47, 53], and recovers the scene flow from it by a linear solve over each locally rigid neighborhood. Sang et al. [38] instead estimate the time-varying field jointly with an explicit velocity field coupled by a modified level-set equation, so that geometry and motion constrain each other rather than one being derived from the other, and 4Deform [37] extends this coupling to sequences and to larger non-isometric deformations.

Three challenges deserve particular attention. First, correspondence under changing topology remains difficult: invertible deformation maps provide explicit point correspondence but generally preserve topology, while spatiotemporal fields can represent splitting and merging but do not by themselves define correspondence through such events. Second, large deformations and long sequences remain challenging, as geometry and correspondence become harder to maintain as motion grows more complex and extends over time. Third, representational coverage remains narrow: general-purpose 4D implicit models still predominantly rely on OFs and SDFs for closed surfaces, while the open and non-manifold surfaces discussed in Section 7.1 of the main paper remain largely unexplored in 4D.

A.3 Applications in Other Domains

Implicit geometric representations support spatial computing applications such as digital twins, heritage preservation, and world models. The reconstruction methods reviewed above turn images and point clouds into 3D surfaces without prescribing mesh connectivity. Neuralangelo [23] recovers fine surface detail from multi-view RGB images, while BakedSDF [52] reconstructs SDF geometry and equips the extracted mesh with view-dependent appearance for photorealistic real-time rendering. DigitalTwinArt [49] combines SDF reconstruction with estimation of joint parameters to build articulated digital twins from RGB-D observations at two articulation states. For heritage documentation, NeuralRecon-W [41] reconstructs historical landmarks from Internet

photographs. World models such as UnO [1] and DIO [10] predict continuous occupancy fields over space and time from LiDAR observations, completing partially observed scenes and forecasting future occupancy.

In medical imaging, implicit fields support anatomical reconstruction and synthesis. NIHCs [27] combine occupancy prediction with anatomical coordinate mapping to reconstruct cardiac anatomy from segmentations of sparse magnetic resonance imaging (MRI) slices. SDF4CHD [19] generates cardiac anatomies with abnormal openings between chambers, while De Wilde et al. [9] synthesize thyroid shapes with connected or separate lobes. These topology variations cannot be captured by deforming a mesh with fixed connectivity.

In robotics, SDF values and gradients provide distance-based objectives and update directions for pose estimation and motion optimization. PIN-SLAM [30] estimates sensor poses by minimizing SDF residuals at incoming LiDAR points, without explicitly matching them to surface points. RDF [22] represents the rigid parts of a robot as SDFs and uses distance and gradient queries to optimize robot configurations for collision avoidance and whole-body manipulation.

In physical simulation, implicit fields provide geometric information directly to numerical solvers. Simplicits [26] can use SDF signs to select interior samples for evaluating elastic energies, enabling simulation without a volumetric mesh. Daviet et al. [8] differentiate elastic responses with respect to implicit geometry to optimize shape and topology for mechanical performance, without repeatedly generating a boundary-conforming volume mesh. For fluid flow, INR-SBM [16] uses neural SDF values and gradients to compute offsets between a grid-aligned boundary and the true surface. These offsets are used to correct wall boundary conditions, enabling simulation on a background grid without a boundary-conforming mesh.

B POINT CLOUD RECONSTRUCTION BENCHMARK

B.1 Evaluation Scenarios

We assemble four evaluation scenarios to cover the diversity of practical reconstruction settings.

Watertight objects. We source 42 shapes from the Famous dataset [12] (22 shapes) and the Google Scanned Objects dataset [11] (20 shapes). Since the raw meshes are not guaranteed to be manifold or self-intersection-free, we apply alpha wrapping [33] to all shapes to produce closed, manifold, self-intersection-free geometry suitable for ground-truth evaluation.

Non-watertight objects. We select 20 shapes that exhibit two common types of topological irregularity. From the non-manifold ABC shapes provided by SALS [36], we take 10 CAD assemblies whose intersecting parts introduce non-manifold edges and vertices, testing a method's ability to handle non-manifold topology. From the MGN dataset [50], we take 10 clothed human body scans that contain open boundaries at garment hems and sleeves, testing reconstruction fidelity on surfaces without closed enclosures.

For each object, five point cloud configurations are generated by uniform surface sampling: three noise-free sets at 3K, 30K, and 100K points (termed *Sparse*, *Clean*, and *Dense*), and two noisy variants that add isotropic Gaussian noise with standard deviation $\sigma \in \{0.002\ell, 0.005\ell\}$ to the 30K configuration, where ℓ denotes the bounding-box diagonal.

Synthetic scenes. We use 5 indoor scenes from the SceneNet dataset, preprocessed following POCO [3] to ensure watertight ground-truth meshes, for evaluating scene-level reconstruction under controlled, noise-free conditions.

Real-scanned scenes. We adopt 5 scenes from the 3D Scene dataset [50], which retain real acquisition artifacts such as sensor noise, to evaluate robustness under practical scanning conditions.

Scene point clouds are sampled at area-based densities of 500 and 1000 points per m^2. In total, the benchmark comprises $42{\times}5 + 20{\times}5 + 5{\times}2 + 5{\times}2 = 330$ evaluation instances.

B.2 Evaluated Methods

We evaluate 19 representative methods covering analytical and neural implicit fields. For *analytical fields*, we include SPSR [17] and SAP [32] as baselines. The remaining 17 methods are *neural fields*, divided into two paradigms.

Learning-based prediction: ConvONet [31], POCO [3], NDF [6], GIFS [54], GeoUDF [34], and SALS [36].

Per-instance fitting, grouped by supervision strategy: direct field supervision (IGR [14], SIREN [39], DiGS [2], NSH [48], StEik [51], S^2DF [50]), projection-based supervision (Neural-Pull [24], I-Filtering [21], LevelSetUDF [57], CAP-UDF [56]), and PG-SDF [18] which combines both.

SPSR and SIREN use oriented normals in our benchmark. IGR is evaluated with both oriented and unoriented inputs to examine the effect of orientation information; the remaining configurations use unoriented inputs.

Among all methods, NDF, GIFS, GeoUDF, SALS, LevelSetUDF, CAP-UDF, and S^2DF support both open- and closed-surface reconstruction; the remaining methods inherently produce closed surfaces and are therefore evaluated only on watertight objects and synthetic scenes. Note that I-Filtering also provides a UDF formulation; we adopt its SDF variant in this benchmark and accordingly include it only in the watertight and synthetic-scene evaluations.

SAP is evaluated in its generalization variant (SAP-Gen) on objects and its optimization variant (SAP-Opt) on scenes, as SAP-Gen generalizes poorly to large-scale scenes while SAP-Opt underperforms SAP-Gen on objects. ConvONet, Neural-Pull, and GeoUDF are excluded from scene-level evaluation, as their time or memory costs become prohibitive with increasing scene extent on a single GPU.

B.3 Implementation and Evaluation Protocol

To facilitate fair comparison, we evaluate all benchmarked methods through a unified evaluation pipeline. Methods are compared on the same benchmark instances within their supported scenarios, with oriented and unoriented inputs reported separately. Each method is wrapped behind the same point-cloud-to-mesh interface, with dependencies isolated in separate micromamba environments. A companion script evaluates all reconstructed meshes in batch and automatically populates the result tables. The benchmark code, evaluation scripts, and configuration files will be made publicly available.

All experiments are conducted on the same server running Ubuntu 18.04.5, equipped with four NVIDIA RTX A6000 GPUs with 48 GiB of memory each and two AMD EPYC 7542 processors. Each GPU-based reconstruction uses a single GPU, while independent instances can run on different GPUs. SPSR runs on the CPU with 32 threads per instance. The Python, PyTorch, and CUDA versions used by each method are listed in Table 1.

ConvONet and POCO use 3K input points for object reconstruction, and POCO uses 10K for scene reconstruction. These points are obtained by spatially uniform subsampling of the input cloud for the corresponding condition, including its noise, rather than from a separate clean reference.

Learning-based methods and SAP-Gen are run with fixed pretrained checkpoints; Table 2 lists the corresponding model identifiers used in our benchmark.

We have verified that none of the pretraining datasets contains the benchmark test instances. However, the models differ in their training data, supervision, and learned priors. Their results should therefore be interpreted as end-to-end reconstruction performance of the evaluated pretrained models, rather than as a controlled comparison of network architectures alone.

Table 1. Software environments used in the benchmark.

Methods	Python	PyTorch	CUDA
IGR, ConvONet, SAP-Gen/Opt, POCO, Neural-Pull, NSH, I-Filtering, GeoUDF, SALS, LevelSetUDF, CAP-UDF	3.8.20	1.13.0	11.6
DiGS, NDF, GIFS	3.9.23	1.8.0	11.1
SIREN, S^2DF	3.8.20	2.0.1	11.7
StEik	3.9.12	1.13.1	11.7
PG-SDF	3.8.20	2.1.2	11.8
SPSR (CPU executable)	–	–	–

Table 2. Pretrained models used by learning-based and generalization-based methods.

Method	Pretrained model
ConvONet (object) [31]	`shapenet_3plane`
SAP-Gen (object) [32]	`ours_noise_005`
POCO (object) [3]	`ShapeNet_None_FKAConv_InterpAttentionKHeadsNet_None`
POCO (scene) [3]	`SyntheticRooms_None_FKAConv_InterpAttentionKHeadsNet_None`
NDF [6]	`shapenet_cars_pretrained`
GIFS [54]	`checkpoint_44h_0m_56s_158456.50156092644`
GeoUDF [34]	`log_reconstruction_100.00_1.000_0.100`
SALS [36]	`logs_thingi10k/01_24_23_02_00`

For per-instance fitting methods, we standardize the optimization budget by supervision type: direct field supervision methods run for 10K gradient steps on objects and 50K on scenes, projection-based methods run for 40K and 80K steps, respectively, and PG-SDF follows its default progressive schedule. SAP-Opt uses 1,600 optimization steps. These shared iteration budgets do not imply equal computational costs across methods.

We use a common resolution setting of 256 for mesh extraction. All open-surface methods except SALS use DCUDF [15] for surface extraction, while other methods follow their default extraction pipelines.

For evaluation, all benchmark instances are preprocessed into a normalized coordinate frame. For real-scanned scenes, predicted faces farther than 0.008 from the input scan are removed before sampling. We uniformly sample 10^5 surface points from both the predicted and ground-truth meshes using fixed random seeds, and normalize both point sets using the center and maximum side length of the ground-truth surface samples, yielding a common unit-scale frame. We report four surface-fidelity metrics:

Chamfer Distance (CD, ↓, L1, $\times 10^3$) measures the average surface deviation between the prediction and the ground truth. For each predicted point we find its nearest neighbor on the ground-truth surface and record the unsquared Euclidean distance, and vice versa. The reported value is the sum of the two resulting means.

F-Score (F1, ↑, %, $\tau = 0.004$) captures the fraction of surface that is accurately reconstructed. A predicted point is counted as a true positive if its nearest ground-truth neighbor is closer than τ, yielding precision; a ground-truth point is counted as recovered if its nearest predicted neighbor falls within the same threshold, yielding recall. We report the harmonic mean of the two.

Table 3. Quantitative results on **watertight object** reconstruction. The three blocks correspond to oriented-input, unoriented closed-surface, and unoriented open-surface methods. Within each block, **bold** and underline mark the best and second-best results, respectively.

		Sparse (3K pts)				Clean (30K pts)				Dense (100K pts)				Noise σ=0.002ℓ				Noise σ=0.005ℓ				Overall Efficiency		
Method	Input	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	T_{field} (s)	T_{extract} (s)	VRAM (GiB)
SPSR [17]	Ori	10.92	62.66	89.64	24.75	5.149	86.25	96.46	6.176	7.335	87.42	97.76	19.85	**5.554**	**83.84**	94.33	**6.112**	**6.839**	71.46	90.19	**7.530**	61.14	16.56	–
IGR [14]	Ori	**6.444**	**79.52**	**94.06**	**11.28**	**4.982**	**87.88**	**97.20**	**6.126**	**4.775**	**88.67**	**98.03**	**4.916**	11.69	83.75	**94.64**	15.17	14.34	**72.70**	**90.51**	22.35	653.11	20.42	4.70
SIREN [39]	Ori	9.412	70.94	93.96	31.89	9.028	78.64	96.38	36.46	9.682	78.96	96.34	27.28	7.207	74.03	94.44	16.13	13.06	47.04	74.27	32.90	266.60	9.64	2.63
ConvONet [31]	Unori	28.30	22.37	77.92	61.76	34.86	20.97	77.80	84.81	35.74	20.06	77.36	87.77	36.04	21.06	77.81	86.33	42.96	8.900	77.30	92.71	21.37	5.32	1.74
IGR [14]	Unori	14.15	66.56	88.28	54.24	12.14	78.23	93.08	45.30	11.42	79.66	93.66	44.26	13.25	74.93	91.72	55.08	28.54	43.63	79.73	73.76	514.62	22.30	5.50
SAP (Gen) [32]	Unori	8.940	57.40	88.72	**12.97**	8.941	57.48	88.69	13.17	8.922	57.43	88.73	13.04	9.084	56.51	88.42	13.25	10.84	46.57	85.76	16.86	8.40	1.80	2.88
POCO [3]	Unori	8.817	59.91	89.49	13.82	8.809	59.63	89.41	13.75	8.847	59.81	89.44	14.21	8.949	58.87	89.25	14.18	10.44	47.96	**86.92**	15.37	25.62	59.45	10.05
Neural-Pull [24]	Unori	9.705	69.11	89.12	23.72	6.651	86.51	96.17	15.80	5.674	87.87	96.83	11.11	**6.073**	**81.22**	92.69	8.089	10.08	42.95	75.06	12.71	758.53	4.22	8.51
DiGS [2]	Unori	8.559	70.12	90.15	17.54	6.480	85.72	96.45	14.81	6.462	85.79	96.70	15.87	7.654	74.65	91.42	16.86	14.40	26.11	57.33	24.10	580.62	12.59	5.01
NSH [48]	Unori	10.57	64.67	88.78	28.52	8.995	70.75	92.73	21.43	10.26	69.11	91.48	25.19	12.83	50.03	82.58	28.95	17.30	24.13	77.91	31.31	1571.58	22.73	6.39
StEik [51]	Unori	9.039	66.46	88.41	17.24	6.134	84.07	95.96	13.09	6.044	85.59	96.75	10.22	7.606	72.61	88.96	12.47	13.16	29.57	57.59	23.29	553.64	22.84	2.46
PG-SDF [18]	Unori	19.27	29.89	74.79	36.89	**5.577**	83.81	94.30	**7.027**	5.101	86.44	95.33	6.162	6.414	76.84	90.07	**7.463**	**8.778**	**53.81**	82.18	**10.57**	446.81	4.31	3.40
I-Filtering [21]	Unori	**7.206**	**77.14**	**92.54**	16.12	5.905	**87.43**	**96.79**	13.72	**4.792**	**88.83**	**97.56**	**5.653**	6.339	80.85	**92.81**	11.18	9.761	44.90	75.74	12.41	2002.33	4.51	2.03
NDF [6]	Unori	11.15	50.18	86.42	26.58	8.840	61.56	93.34	20.98	9.160	59.37	93.66	22.30	10.87	42.72	91.68	23.06	22.09	3.810	**87.22**	21.76	5.07	345.36	20.90
GIFS [54]	Unori	9.724	56.23	85.45	17.28	7.289	68.08	93.48	9.981	8.183	56.89	93.89	10.82	11.05	28.58	91.34	13.01	22.10	4.640	83.07	22.09	13.36	169.40	9.86
GeoUDF [34]	Unori	**6.709**	**74.51**	89.74	**9.657**	4.855	87.62	96.02	**5.130**	4.736	88.44	96.80	5.000	8.119	53.88	89.50	8.547	18.39	7.510	79.31	19.39	10.24	1158.33	14.44
LevelSetUDF [57]	Unori	8.569	66.64	87.03	15.89	4.895	88.11	96.51	6.205	4.785	88.52	96.90	5.242	6.312	76.10	92.23	**6.544**	13.46	21.56	83.44	15.87	1126.89	234.02	8.51
CAP-UDF [56]	Unori	11.66	62.04	84.36	23.95	4.995	87.67	96.27	6.076	4.917	87.98	96.63	5.853	6.337	75.76	90.91	6.590	11.91	29.72	73.34	15.11	713.57	243.81	8.51
SALS [36]	Unori	8.408	65.69	77.43	14.93	5.123	85.71	92.81	6.362	4.807	87.90	95.17	5.180	7.507	63.13	72.61	9.608	12.44	28.74	58.70	17.12	183.65	25.46	19.56
S^2DF [50]	Unori	14.93	41.52	**89.88**	32.04	**4.776**	**88.34**	**97.09**	5.372	**4.642**	**89.22**	**97.82**	**4.862**	**6.311**	**76.67**	**92.94**	6.741	**9.326**	**44.44**	86.40	**10.73**	760.51	623.30	15.54

Normal Consistency (NC, ↑, %) measures agreement between predicted and ground-truth surface normals. For each nearest-neighbor pair identified in the Chamfer Distance computation, we take the absolute cosine similarity between their normals. The reported value is the average across both matching directions.

Hausdorff Distance (HD, ↓, 97th-percentile, $\times 10^3$) provides a robust measure of large surface deviations. In each matching direction we compute the 97th percentile of the nearest-neighbor distances, and report the larger of the two.

For efficiency evaluation, we measure wall-clock time separately for point-cloud-to-field construction and field-to-mesh extraction. Five warm-up runs per GPU are excluded, and intermediate mesh extraction is disabled. GPU memory usage is sampled every 250 ms for the reconstruction process and its subprocesses.

For each method and scenario, runtimes are averaged within each condition and then across conditions. VRAM is the maximum observed GPU memory usage across the measured runs.

B.4 Quantitative Reconstruction Results

Table 4. Quantitative results on **non-watertight object** reconstruction. Only open-surface methods are evaluated. **Bold** and underline mark the best and second-best results, respectively.

	Sparse (3K pts)				Clean (30K pts)				Dense (100K pts)				Noise σ=0.002ℓ				Noise σ=0.005ℓ				Overall Efficiency		
Method	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	T_{field} (s)	T_{extract} (s)	VRAM (GiB)
NDF [6]	8.726	62.73	91.13	21.18	7.293	71.42	95.88	18.53	8.160	64.00	95.06	21.61	9.019	49.89	94.57	10.87	21.40	3.000	89.80	20.41	4.77	273.64	19.86
GIFS [54]	7.669	69.25	89.35	13.32	6.514	74.70	95.25	7.690	7.949	55.90	93.34	9.888	10.26	28.96	93.52	9.993	21.46	3.400	86.26	19.20	14.20	151.96	9.57
GeoUDF [34]	**5.248**	**84.23**	**93.60**	**7.678**	4.623	90.67	97.26	11.50	4.536	90.77	97.14	5.058	7.233	64.75	91.37	7.438	18.10	6.130	81.96	18.54	12.27	1087.60	14.44
LevelSetUDF [57]	7.143	73.38	90.36	16.67	4.090	90.86	96.58	4.344	4.726	90.27	96.59	11.05	**5.545**	**82.92**	94.46	**5.390**	13.15	19.21	86.45	13.98	1125.25	185.28	8.51
CAP-UDF [56]	8.504	67.51	85.07	20.09	5.116	89.46	96.52	13.87	5.386	88.89	95.84	18.00	9.478	79.11	92.52	20.99	17.31	25.87	76.63	37.16	715.11	224.68	8.51
SALS [36]	6.637	76.34	82.77	11.96	4.314	89.52	94.05	5.163	4.108	90.97	95.84	4.407	6.546	72.73	72.28	8.042	11.31	35.36	58.18	16.01	70.46	2.20	17.98
S^2DF [50]	11.30	57.17	92.53	25.28	**4.004**	**91.59**	**98.24**	**4.337**	**3.971**	**91.82**	**98.43**	**4.224**	5.724	81.62	**95.26**	6.082	**8.478**	**54.21**	**90.69**	**9.779**	755.72	285.63	5.00

Tables 3, 4, and 5 report reconstruction accuracy and computational cost for watertight objects, non-watertight objects, and scenes, respectively, following the evaluation protocol in Section B.3.

Table 5. Quantitative results on **synthetic and real-scanned scene** reconstruction. For synthetic scenes, methods are divided into oriented-input, unoriented closed-surface, and unoriented open-surface blocks; for real-scanned scenes, only open-surface methods are evaluated. Within each block, **bold** and underline mark the best and second-best results, respectively.

		500 pts/m^2				1000 pts/m^2				Overall Efficiency		
Method	Input	CD↓	F1↑	NC↑	HD↓	CD↓	F1↑	NC↑	HD↓	T_{field} (s)	T_{extract} (s)	VRAM (GiB)
Synthetic scene												
SPSR [17]	Ori	**7.231**	**62.14**	**96.32**	**7.547**	**7.143**	**62.66**	**96.76**	**7.418**	86.54	36.10	–
IGR [14]	Ori	8.684	58.35	95.81	16.33	7.755	59.48	95.94	10.21	4049.93	64.57	5.48
SIREN [39]	Ori	8.118	53.16	96.23	7.652	8.308	52.36	96.19	8.434	855.49	11.55	2.63
IGR [14]	Unori	17.33	38.61	81.12	51.09	15.25	39.84	86.34	45.03	3228.26	83.32	6.33
SAP (Opt) [32]	Unori	9.223	53.22	81.54	19.44	**8.180**	**54.67**	84.10	**9.047**	1694.02	1.75	3.44
POCO [3]	Unori	9.190	50.58	92.03	14.46	9.299	53.41	91.63	18.33	14.39	109.85	9.37
DiGS [2]	Unori	8.706	54.32	94.41	14.92	8.705	54.19	**94.64**	14.39	9442.99	39.37	13.23
NSH [48]	Unori	19.16	39.69	86.77	46.89	10.41	50.43	91.90	22.99	4696.42	27.73	6.39
StEik [51]	Unori	**7.939**	**56.15**	**95.43**	**8.612**	8.404	54.26	94.56	10.76	7267.30	60.36	5.62
PG-SDF [18]	Unori	9.607	52.18	87.97	17.94	9.505	52.67	87.93	17.21	1518.23	5.14	16.54
I-Filtering [21]	Unori	9.270	55.38	92.29	16.29	10.61	53.38	90.75	27.02	6873.29	72.59	4.53
NDF [6]	Unori	16.15	30.41	89.25	80.03	18.98	23.67	87.17	86.96	5.83	498.26	19.71
GIFS [54]	Unori	9.301	44.04	91.95	9.883	9.891	40.46	91.33	11.64	15.29	262.31	9.42
LevelSetUDF [57]	Unori	7.996	58.54	93.79	9.155	7.808	58.86	93.98	8.520	9337.62	701.83	11.18
CAP-UDF [56]	Unori	7.744	58.70	93.90	8.417	7.881	58.81	93.96	8.676	1497.20	719.51	8.36
SALS [36]	Unori	7.830	56.66	85.43	8.255	7.831	56.66	85.47	8.271	174.61	2.81	16.40
S^{2}DF [50]	Unori	**7.283**	**61.45**	**95.67**	**7.813**	**7.290**	**61.36**	**95.77**	**7.837**	4275.91	2036.49	8.22
Real-scanned scene												
NDF [6]	Unori	6.885	67.16	91.08	7.439	7.059	62.14	90.55	8.274	4.56	172.91	19.86
GIFS [54]	Unori	6.889	71.78	90.37	8.239	7.874	57.87	88.40	8.446	13.06	78.88	10.04
LevelSetUDF [57]	Unori	3.685	95.74	92.62	4.084	3.784	94.79	92.36	4.626	2269.50	168.81	6.70
CAP-UDF [56]	Unori	3.805	95.31	92.07	4.255	3.852	94.66	92.11	4.684	1560.30	216.91	6.70
SALS [36]	Unori	3.952	93.78	87.95	4.766	3.880	94.22	88.92	4.637	71.86	1.93	14.78
S^{2}DF [50]	Unori	**3.599**	**96.16**	**93.44**	**3.990**	**3.628**	**95.74**	**93.79**	**4.171**	4270.83	278.61	4.58

B.5 Mesh-quality case study

We evaluate mesh quality for all methods on the watertight-object benchmark to complement surface-fidelity metrics (Table 6). We count non-manifold vertices (NM-V), whose incident faces do not form a single fan, and non-manifold edges (NM-E), which are shared by more than two faces. We also count self-intersecting faces (SI-F) and vertex-connected components (CC).

Triangle quality is measured by the normalized mean ratio [40], defined as $q(f) = 4\sqrt{3}A_f/(a_f^2 + b_f^2 + c_f^2)$, where A_f is the triangle area and a_f, b_f, c_f are its edge lengths. The score lies in $[0, 1]$, with 1 for equilateral triangles and 0 assigned to degenerate triangles. We average the per-mesh means and medians to obtain Q_{mean} and Q_{median}. Faces with repeated vertex indices are excluded from topology checks but contribute zero to triangle quality.

In our benchmark, all UDF-based methods except SALS use DCUDF [15], whereas SDF-based methods generally use their native Marching Cubes implementations. The choice of surface extraction algorithm is therefore a key factor in the reported mesh quality, and these results should not be attributed solely to the point-cloud reconstruction methods.

Table 6. Mesh quality on **watertight objects**, averaged over meshes from all five input conditions. The three blocks contain oriented-input, unoriented closed-surface, and unoriented open-surface methods.

Method	Input	NM-V↓	NM-E↓	SI-F↓	CC	Q_{mean} ↑	Q_{median} ↑
SPSR [17]	Ori	0	0	0.019	2.75	0.719	0.835
IGR [14]	Ori	0.210	0.105	0	59.92	0.701	0.830
SIREN [39]	Ori	0.100	0.048	0	304.20	0.695	0.825
ConvONet [31]	Unori	0	0	0.0095	3.84	0.709	0.843
IGR [14]	Unori	0.081	0.029	0	106.80	0.700	0.830
SAP (Gen) [32]	Unori	0	0	0	18.49	0.705	0.837
POCO [3]	Unori	0	0	2.19	15.66	0.705	0.836
Neural-Pull [24]	Unori	0	0	0	45.41	0.667	0.768
DiGS [2]	Unori	0.467	0.229	0	750.81	0.694	0.827
NSH [48]	Unori	0.100	0.029	0.248	147.66	0.693	0.827
StEik [51]	Unori	0.201	0.067	0.191	26.59	0.693	0.826
PG-SDF [18]	Unori	0	0	0.071	90.75	0.690	0.823
I-Filtering [21]	Unori	0.0095	0.0048	0	80.22	0.667	0.767
NDF [6]	Unori	0	0	1.72×10^{4}	6.40	0.861	0.895
GIFS [54]	Unori	0.024	0.0095	2.18×10^{4}	114.04	0.857	0.893
GeoUDF [34]	Unori	0.033	0.024	1.97×10^{5}	285.75	0.854	0.892
LevelSetUDF [57]	Unori	0	0	5.17×10^{5}	18.27	0.851	0.891
CAP-UDF [56]	Unori	0	0	6.04×10^{5}	50.38	0.839	0.882
SALS [36]	Unori	1.95×10^{5}	2.29×10^{5}	2.67×10^{5}	116.38	0.596	0.657
S^2DF [50]	Unori	0.043	0.014	5.38×10^{5}	154.40	0.804	0.833

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